



Module Manual

Master of Science (M.Sc.)

Computer Science

Cohort: Winter Term 2025

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Program description

Content

Core Qualification

Module M0523: Business & Management

Module Responsible	Prof. Matthias Meyer
Admission Requirements	Successful completion of the modul "Foundations of Management"
Recommended Previous Knowledge	None
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence <i>Knowledge</i> - Students are able to find their way around selected special areas of management within the scope of business management. - Students are able to explain basic theories, categories, and models in selected special areas of business management. - Students are able to interrelate technical and management knowledge. <i>Skills</i> - Students are able to apply basic methods in selected areas of business management. - Students are able to explain and give reasons for decision proposals on practical issues in areas of business management. Personal Competence <i>Social Competence</i> - Students are able to communicate in small interdisciplinary groups and to jointly develop solutions for complex problems <i>Autonomy</i> - Students are capable of acquiring necessary knowledge independently by means of research and preparation of material.	
Workload in Hours	Depends on choice of courses
Credit points	6

Courses

Information regarding lectures and courses can be found in the corresponding module handbook published separately.

Module M0524: Non-technical Courses for Master	
Module Responsible	Dagmar Richter
Admission Requirements	None
Recommended Previous Knowledge	None
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence <i>Knowledge</i>	The Nontechnical Academic Programms (NTA) imparts skills that, in view of the TUHH's training profile, professional engineering studies require but are not able to cover fully. Self-reliance, self-management, collaboration and professional and personnel management competences. The department implements these training objectives in its teaching architecture, in its teaching and learning arrangements, in teaching areas and by means of teaching offerings in which students can qualify by opting for specific competences and a competence level at the Bachelor's or Master's level. The teaching offerings are pooled in two different catalogues for nontechnical complementary courses. The Learning Architecture consists of a cross-disciplinarily study offering. The centrally designed teaching offering ensures that courses in the nontechnical academic programms follow the specific profiling of TUHH degree courses. The learning architecture demands and trains independent educational planning as regards the individual development of competences. It also provides orientation knowledge in the form of "profiles". The subjects that can be studied in parallel throughout the student's entire study program - if need be, it can be studied in one to two semesters. In view of the adaptation problems that individuals commonly face in their first semesters after making the transition from school to university and in order to encourage individually planned semesters abroad, there is no obligation to study these subjects in one or two specific semesters during the course of studies. Teaching and Learning Arrangements provide for students, separated into B.Sc. and M.Sc., to learn with and from each other across semesters. The challenge of dealing with interdisciplinarity and a variety of stages of learning in courses are part of the learning architecture and are deliberately encouraged in specific courses. Fields of Teaching are based on research findings from the academic disciplines cultural studies, social studies, arts, historical studies, communication studies, migration studies and sustainability research, and from engineering didactics. In addition, from the winter semester 2014/15 students on all Bachelor's courses will have the opportunity to learn about business management and start-ups in a goal-oriented way. The fields of teaching are augmented by soft skills offers and a foreign language offer. Here, the focus is on encouraging goal-oriented communication skills, e.g. the skills required by outgoing engineers in international and intercultural situations. The Competence Level of the courses offered in this area is different as regards the basic training objective in the Bachelor's and Master's fields. These differences are reflected in the practical examples used, in content topics that refer to different professional application contexts, and in the higher scientific and theoretical level of abstraction in the B.Sc. This is also reflected in the different quality of soft skills, which relate to the different team positions and different group leadership functions of Bachelor's and Master's graduates in their future working life. Specialized Competence (Knowledge) Students can • explain specialized areas in context of the relevant non-technical disciplines, • outline basic theories, categories, terminology, models, concepts or artistic techniques in the disciplines represented in the learning area, • different specialist disciplines relate to their own discipline and differentiate it as well as make connections, • sketch the basic outlines of how scientific disciplines, paradigms, models, instruments, methods and forms of representation in the specialized sciences are subject to individual and socio-cultural interpretation and historicity, • Can communicate in a foreign language in a manner appropriate to the subject.
<i>Skills</i>	Professional Competence (Skills) In selected sub-areas students can • apply basic and specific methods of the said scientific disciplines, • question a specific technical phenomena, models, theories from the viewpoint of another, aforementioned specialist discipline, • to handle simple and advanced questions in aforementioned scientific disciplines in a successful manner, • justify their decisions on forms of organization and application in practical questions in contexts that go beyond the technical relationship to the subject.
Personal Competence <i>Social Competence</i>	Personal Competences (Social Skills)

<i>Autonomy</i>	Students will be able • to learn to collaborate in different manner, • to present and analyze problems in the abovementioned fields in a partner or group situation in a manner appropriate to the addressees, • to express themselves competently, in a culturally appropriate and gender-sensitive manner in the language of the country (as far as this study-focus would be chosen), • to explain nontechnical items to auditorium with technical background knowledge. Personal Competences (Self-reliance) Students are able in selected areas • to reflect on their own profession and professionalism in the context of real-life fields of application • to organize themselves and their own learning processes • to reflect and decide questions in front of a broad education background • to communicate a nontechnical item in a competent way in written form or verbally • to organize themselves as an entrepreneurial subject country (as far as this study-focus would be chosen)
Workload in Hours	Depends on choice of courses
Credit points	6

Courses
Information regarding lectures and courses can be found in the corresponding module handbook published separately.

Module M1563: Research Project Computer Science								
Courses								
Title		Typ	Hrs/wk	CP				
Research Project Computer Science (L2353)		Projection Course	8	12				
Module Responsible	Dozenten des SD E							
Admission Requirements	None							
Recommended Previous Knowledge	Basic knowledge and techniques from the Master courses in the semesters 1 and 2.							
Educational Objectives	After taking part successfully, students have reached the following learning results							
Professional Competence	Knowledge Students are able to acquire advanced knowledge in a subfield of Computer Science and can independently acquire deeper knowledge in the field. Skills The students are able to formulate the scientific problems to be considered and to work out solutions in an independent manner and to realize them. Personal Competence Social Competence The students are able to discuss proposals for solutions of scientific problems within the team. They are able to present the results in a clear and well structured manner. Autonomy The students can provide a scientific work in a timely manner and document the results in a detailed and well readable form. They are able to actively follow anticipate the presentations of other students such that eventually a scientific discussion comes up.							
Workload in Hours					Independent Study Time 248, Study Time in Lecture 112			
Credit points					12			
Course achievement					None			
Examination	Study work							
Examination duration and scale	Vortrag							
Assignment for the Following Curricula	Computer Science: Core Qualification: Compulsory Data Science: Core Qualification: Compulsory							
Course L2353: Research Project Computer Science								
Typ	Projection Course							
Hrs/wk	8							
CP	12							
Workload in Hours	Independent Study Time 248, Study Time in Lecture 112							
Lecturer	Dozenten des SD E							
Language	EN							
Cycle	WiSe							
Content	Current research topics of the chosen areas of specialization							
Literature	Wird vom Veranstalter bekanntgegeben.							

Specialization I. Computer and Software Engineering

Module M0753: Software Verification

Courses

Title	Typ	Hrs/wk	CP
Software Verification (L0629)	Lecture	2	3
Software Verification (L0630)	Recitation Section (small)	2	3

Module Responsible	Prof. Sibylle Schupp
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Admission Requirements	None
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Recommended Previous Knowledge	- Automata theory and formal languages - Computational logic - Object-oriented programming, algorithms, and data structures - Functional programming or procedural programming - Concurrency
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Educational Objectives	After taking part successfully, students have reached the following learning results
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Professional Competence	
<i>Knowledge</i>	Students apply the major verification techniques in model checking and deductive verification. They explain in formal terms syntax and semantics of the underlying logics, and assess the expressivity of different logics as well as their limitations. They classify formal properties of software systems. They find flaws in formal arguments, arising from modeling artifacts or underspecification.
<i>Skills</i>	Students formulate provable properties of a software system in a formal language. They develop logic-based models that properly abstract from the software under verification and, where necessary, adapt model or property. They construct proofs and property checks by hand or using tools for model checking or deductive verification, and reflect on the scope of the results. Presented with a verification problem in natural language, they select the appropriate verification technique and justify their choice.
Personal Competence	
<i>Social Competence</i>	Students discuss relevant topics in class. They defend their solutions orally. They communicate in English.
<i>Autonomy</i>	Using accompanying on-line material for self study, students can assess their level of knowledge continuously and adjust it appropriately. Working on exercise problems, they receive additional feedback. Within limits, they can set their own learning goals. Upon successful completion, students can identify and precisely formulate new problems in academic or applied research in the field of software verification. Within this field, they can conduct independent studies to acquire the necessary competencies and compile their findings in academic reports. They can devise plans to arrive at new solutions or assess existing ones.

Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
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Credit points	6
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Course achievement	Compulsory	Bonus	Form	Description
	Yes	15 %	Exercises	

Examination	Written exam
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Examination duration and scale	90 min
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Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory
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Course L0629: Software Verification

Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Sibylle Schupp
Language	EN
Cycle	WiSe
Content	- Model checking (bounded model checking, CTL, LTL) - Real-time model checking (TCTL, timed automata) - Deductive verification (Hoare logic) - Tool support - Recent developments of verification techniques and applications
Literature	C. Baier and J-P. Katoen, Principles of Model Checking, MIT Press 2007. M. Huth and M. Bryan, Logic in Computer Science. Modelling and Reasoning about Systems, 2nd Edition, 2004. - Selected Research Papers

Course L0630: Software Verification	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Sibylle Schupp
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0942: Software Security				
Courses				
Title	Typ		Hrs/wk	CP
Software Security (L1103)	Lecture		2	3
Software Security (L1104)	Recitation Section (small)		2	3
Module Responsible	Prof. Riccardo Scandariato			
Admission Requirements	None			
Recommended Previous Knowledge	Familiarity with C/C++, web programming			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	Students can - name the main causes for security vulnerabilities in software - explain current methods for identifying and avoiding security vulnerabilities Students are capable of - performing a software vulnerability analysis - developing secure code			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>	None			
<i>Autonomy</i>	Students are capable of acquiring knowledge independently from professional publications, technical standards, and other sources, and are capable of applying newly acquired knowledge to new problems.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	5 %	Subject theoretical practical work	and Gruppenarbeit mit aktuellen Technologien aus dem Bereich Sicherheit
Examination	Written exam			
Examination duration and scale	120 minutes			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Elective Compulsory			

Course L1103: Software Security	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Riccardo Scandariato
Language	EN
Cycle	WiSe
Content	- Attacks exploiting character and integer representations - Buffer overruns - Vulnerabilities in memory management: use-after-free attacks - SQL injection - Cross-site scripting and cross-site request forgery - Side-channel attacks - Attacks in mobile applications - Testing for security; taint analysis - Type safe languages - Automatic vulnerability repair
Literature	M. Howard, D. LeBlanc: Writing Secure Code, 2nd edition, Microsoft Press (2002) G. Hoglund, G. McGraw: Exploiting Software: How to Break Code, Addison-Wesley (2004)

Course L1104: Software Security	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Riccardo Scandariato
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1694: Security of Cyber-Physical Systems				
Courses				
Title	Typ		Hrs/wk	CP
Security of Cyber-Physical Systems (L2691)	Integrated Lecture		3	4
Security of Cyber-Physical Systems (L2692)	Recitation Section (small)		1	2
Module Responsible	Prof. Sibylle Fröschle			
Admission Requirements	None			
Recommended Previous Knowledge	IT security, programming skills, statistics			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> The students know and can explain - the threats posed by cyber attacks to cyber-physical systems (CPS) - concrete attacks at a technical level, e.g. on bus systems - security solutions specific to CPS with their capabilities and limitations - examples of security architectures for CPS and the requirements they guarantee - standard security engineering processes for CPS <i>Skills</i> The students are able to - identify security threats and assess the risks for a given CPS - apply attack toolkits to analyse a networked control system, and detect attacks beyond those taught in class - identify and apply security solutions suitable to the requirements - follow security engineering processes to develop a security architecture for a given CPS - recognize challenges and limitations, e.g. posed by novel types of attack			
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>				
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	10 %	Exercises	Die Übungsaufgaben finden semesterbegleitend statt.
Examination	Written exam			
Examination duration and scale	120 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory			

Course L2691: Security of Cyber-Physical Systems	
Typ	Integrated Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Sibylle Fröschle
Language	EN
Cycle	WiSe
Content	Embedded systems in energy, production, and transportation are currently undergoing a technological transition to highly networked automated cyber-physical systems (CPS). Such systems are potentially vulnerable to cyber attacks, and these can have physical impact. In this course we investigate security threats, solutions and architectures that are specific to CPS. The topics are as follows: Fundamentals and motivating examples Networked and embedded control systems Bus system level attacks Intruder detection systems (IDS), in particular physics-based IDS System security architectures, including cryptographic solutions Adversarial machine learning attacks in the physical world Aspects of Location and Localization Wireless networks and infrastructures for critical applications Communication security architectures and remaining threats Intruder detection systems (IDS), in particular data-centric IDS Resilience against multi-instance attacks Security Engineering of CPS: Process and Norms
Literature	Recent scientific papers and reports in the public domain.

Course L2692: Security of Cyber-Physical Systems	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Sibylle Fröschle
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1749: Energy Efficiency in Embedded Systems			
Courses			
Title	Typ	Hrs/wk	CP
Energy Efficiency in Embedded Systems (L2870)	Lecture	2	3
Energy Efficiency in Embedded Systems (L2872)	Project/problem-based Learning	2	2
Energy Efficiency in Embedded Systems (L2871)	Recitation Section (large)	1	1
Module Responsible	Prof. Ulf Kulau		
Admission Requirements	None		
Recommended Previous Knowledge	• Computer Engineering (mandatory) • Programming Skills in C (mandatory) • Computer Architecture (recommended) ENG: The following books are very good sources for the necessary background knowledge: 1. Harris, David, and N. Weste: CMOS VLSI Design ed., Pearson Education, 2010 2. Rabaey, Jan: Low Power Design Essentials (Integrated Circuits and Systems), Springer, 2009		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i>	Motivation: In the field of computer science we have only limited possibilities to influence the efficiency of the hardware directly, respectively we are dependent on the manufacturers (e.g. of microcontrollers). However, in order to exploit the full potential of the hardware we are given at the system level, we need a deeper understanding of the background, processes and mechanisms of power dissipation in embedded systems. Where does the power dissipation come from, what happens at the hardware level, what mechanisms can I use directly/indirectly, what is the tradeoff between flexibility and efficiency,..... are only a few questions, which will be elaborated and discussed in this event. Contents of teaching: • Motivation and power dissipation on semiconductor level • Power dissipation of digital circuits, in particular CMOS • Power Management in Hard- and Software (Sleep Modes, DVS, FS, Undervolting) • Energy efficient system design (applications) • Energy Harvesting and Transiently Powered Computing (TPC)		
<i>Skills</i>	Upon completion of this module, students will have a deeper understanding of hardware and software mechanisms for evaluating and developing energy-efficient embedded systems • They have a deeper understanding of the electrotechnical basics of power dissipation in digital systems • They can analyze the power dissipation of systems at any level and apply appropriate methods to increase efficiency • They can use a variety of standard techniques to achieve "Energy Efficiency by Design" • They can model, evaluate as well as implement energy-autonomous systems		
Personal Competence <i>Social Competence</i>	As part of the module, concepts learned in the lecture will be implemented on a hardware platform within small groups. Students learn to work in a team and to develop solutions together. Specific tasks are worked on within the group, whereby cross-group collaboration (exchange) also takes place. The second part is a challenge-based project in which the groups find the most energy-efficient solutions possible in healthy competition with each other. This strengthens the cohesion in the groups and reinforces mutual motivation, support and creativity.		
<i>Autonomy</i>	After completing this module, students will be able to independently develop, optimize and evaluate solutions for embedded systems based on the knowledge they have acquired and further technical literature.		
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	90 min		
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Nanoelectronics and Microsystems Technology: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Wireless and Sensor Technologies: Elective Compulsory Electrical Engineering: Specialisation Wireless and Sensor Technologies: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Signal Processing: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Embedded Systems: Elective Compulsory		

Course L2870: Energy Efficiency in Embedded Systems	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Ulf Kulau
Language	EN
Cycle	WiSe
Content	Motivation: In the field of computer science we have only limited possibilities to influence the efficiency of the hardware directly, respectively we are dependent on the manufacturers (e.g. of microcontrollers). However, in order to exploit the full potential of the hardware we are given at the system level, we need a deeper understanding of the background, processes and mechanisms of power dissipation in embedded systems. Where does the power dissipation come from, what happens at the hardware level, what mechanisms can I use directly/indirectly, what is the tradeoff between flexibility and efficiency,..... are only a few questions, which will be elaborated and discussed in this event. Contents of teaching: • Motivation and power dissipation on semiconductor level • Power dissipation of digital circuits, in particular CMOS • Power Management in Hard- and Software (Sleep Modes, DVS, FS, Undervolting) • Energy efficient system design (applications) • Energy Harvesting and Transiently Powered Computing (TPC)
Literature	DE: Die Vorlesungsinhalte wurden in einem Paper veröffentlicht. All Quellen, welche in diesem Kurs verwendet wurden sind in folgenden Paper aufgelistet: ENG: The lecture contents were published in a paper. All sources used in this course are listed in the following paper: 1. Kulau, Ulf: Course: Energy Efficiency in Embedded Systems-A System-Level Perspective for Computer Scientists, EWME, 2018. DE: Als sehr gute Quellen für notwendiges Hintergrundwissen eignen sich die folgenden Bücher: ENG: The following books are very good sources for the necessary background knowledge: 1. Harris, David, and N. Weste: CMOS VLSI Design ed., Pearson Education, 2010 2. Rabaey, Jan: Low Power Design Essentials (Integrated Circuits and Systems), Springer, 2009

Course L2872: Energy Efficiency in Embedded Systems	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Ulf Kulau
Language	EN
Cycle	WiSe
Content	In this project-based exercise, the learned aspects for achieving energy-efficient embedded systems are implemented and consolidated in practical environments in a small project. First, a tool set for the implementation of energy efficiency mechanisms is implemented in common exercises by means of defined tasks. In the second part, a challenge-based exercise is carried out in which a system that is as efficient as possible is to be implemented independently. A system based on an AVR micro-controller is used, which can be operated autonomously by a Solar-Energy Harvester. 1. Task phase: 6 "hands-on" tasks to gain experience and to create a SW library. 2. Project phase: Implementation of an energy autonomous system with the goal of highest possible energy efficiency (Challenge)
Literature	DE: Die Vorlesungsinhalte wurden in einem Paper veröffentlicht. Alle Quellen, welche in diesem Kurs (inklusive Übungen und PBL) verwendet wurden sind in folgenden Paper aufgelistet: ENG: The lecture contents were published in a paper. All sources used in this course (including exercises and PBL) are listed in the following paper: 1. Kulau, Ulf: Course: Energy Efficiency in Embedded Systems-A System-Level Perspective for Computer Scientists, EWME, 2018. DE: Als sehr gute Quellen für notwendiges Hintergrundwissen eignen sich die folgenden Bücher: ENG: The following books are very good sources for the necessary background knowledge: 1. Harris, David, and N. Weste: CMOS VLSI Design ed., Pearson Education, 2010 2. Rabaey, Jan: Low Power Design Essentials (Integrated Circuits and Systems), Springer, 2009

Course L2871: Energy Efficiency in Embedded Systems	
Typ	Recitation Section (large)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Ulf Kulau
Language	EN
Cycle	WiSe
Content	In the lecture hall exercise, the theoretical basics taught in the lecture are deepened. This is done through in-depth discussion of relevant aspects, but also through calculation examples, in which a deeper understanding of the topic of energy efficiency in embedded systems is gained. Exercises will be distributed in advance and solutions will be presented in the lecture hall exercise. Contents of the exercise are as follows: • Basics and calculation of power dissipation on semiconductor • Power dissipation of CMOS using the example of an inverter • Influence of the activity factor and external components • DVS and scheduling • Evaluation to show the benefit of undervolting • Aspects of energy harvesting (MPPT) •
Literature	DE: Die Vorlesungsinhalte wurden in einem Paper veröffentlicht. Alle Quellen, welche in diesem Kurs (also auch den Übungen) verwendet wurden sind in folgenden Paper aufgelistet: ENG: The lecture contents were published in a paper. All sources used in this course (including exercises, and PBL) are listed in the following paper: 1. Kulau, Ulf: Course: Energy Efficiency in Embedded Systems-A System-Level Perspective for Computer Scientists, EWME, 2018. DE: Als sehr gute Quellen für notwendiges Hintergrundwissen eignen sich die folgenden Bücher: ENG: The following books are very good sources for the necessary background knowledge: 1. Harris, David, and N. Weste: CMOS VLSI Design ed., Pearson Education, 2010 2. Rabaey, Jan: Low Power Design Essentials (Integrated Circuits and Systems), Springer, 2009

Module M1875: Algorithm Engineering				
Courses				
Title	Typ		Hrs/wk	CP
Algorithm Engineering (L3089)	Lecture		2	3
Algorithm Engineering (L3090)	Recitation Section (small)		2	3
Module Responsible	Prof. Matthias Mnich			
Admission Requirements	None			
Recommended Previous Knowledge	• Programming skills (C++ or Python) • Algorithms and Data Structures • Basic probability theory • Graph Theory and Optimisation			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i> • Students will learn the best practices in algorithm engineering. • Students will learn the best practical methods to solve important combinatorial optimisation problems. • Students will learn how to use a rigorous methodology to empirically evaluate the performance and accuracy of implemented algorithms. <i>Skills</i> • Students will know how to model computational real-world problems. • Students will know how to transform theoretical designs into efficient programs. • Students will know how to analyse the key performance properties of algorithms. Personal Competence <i>Social Competence</i> • Students will know how to work in groups to solve advanced algorithmic problems. • Students will be capable of implementing separate parts of an algorithm and combining them in a unified way. <i>Autonomy</i> • Students will know how to combine theory and implementation. • Students will learn how to reason about solving an advanced algorithmic problem.				
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Oral exam			
Examination duration and scale	30 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory			

Course L3089: Algorithm Engineering	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	NN
Language	EN
Cycle	WiSe
Content	
Literature	

Course L3090: Algorithm Engineering	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	NN
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1400: Design of Dependable Systems				
Courses				
Title	Typ		Hrs/wk	CP
Designing Dependable Systems (L2000)	Lecture		2	3
Designing Dependable Systems (L2001)	Recitation Section (small)		2	3
Module Responsible	Prof. Görschwin Fey			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge about data structures and algorithms			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	In the following "dependable" summarizes the concepts Reliability, Availability, Maintainability, Safety and Security. Knowledge about approaches for designing dependable systems, e.g., • Structural solutions like modular redundancy • Algorithmic solutions like handling byzantine faults or checkpointing Knowledge about methods for the analysis of dependable systems <i>Skills</i> Ability to implement dependable systems using the above approaches. Ability to analyze the dependability of systems using the above methods for analysis. Personal Competence <i>Social Competence</i> Students • discuss relevant topics in class and • present their solutions orally. <i>Autonomy</i> Using accompanying material students independently learn in-depth relations between concepts explained in the lecture and additional solution strategies.			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>				
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	None	Subject theoretical and practical work	Die Lösung einer Aufgabe ist Zulassungsvoraussetzung für die Prüfung. Die Aufgabe wird in Vorlesung und Übung definiert.
Examination	Oral exam			
Examination duration and scale	30 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Embedded Systems: Elective Compulsory			

Course L2000: Designing Dependable Systems	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Görschwin Fey
Language	EN
Cycle	SoSe
Content	Description The term dependability comprises various aspects of a system. These are typically: • Reliability • Availability • Maintainability • Safety • Security This makes dependability a core aspect that has to be considered early in system design, no matter whether software, embedded systems or full scale cyber-physical systems are considered. Contents The module introduces the basic concepts for the design and the analysis of dependable systems. Design examples for getting practical hands-on-experience in dependable design techniques. The module focuses towards embedded systems. The following topics are covered: • Modelling • Fault Tolerance • Design Concepts • Analysis Techniques
Literature	The lecture is based on a selection of research papers that is regularly updated. The specific literature references are provided with the lecture slides.

Course L2001: Designing Dependable Systems	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Görschwin Fey
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1397: Model Checking - Proof Engines and Algorithms				
Courses				
Title			Typ	Hrs/wk
Model Checking - Proof Engines and Algorithms (L1979)			Lecture	2
Model Checking - Proof Engines and Algorithms (L1980)			Recitation Section (small)	2
Module Responsible	Prof. Görschwin Fey			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge about data structures and algorithms			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	Students know - algorithms and data structures for model checking, - basics of Boolean reasoning engines and - the impact of specification and modelling on the computational effort for model checking.			
<i>Skills</i>	Students can - explain and implement algorithms and data structures for model checking, - decide whether a given problem can be solved using Boolean reasoning or model checking, and - implement the respective algorithms.			
Personal Competence				
<i>Social Competence</i>	Students - discuss relevant topics in class and - defend their solutions orally.			
<i>Autonomy</i>	Using accompanying material students independently learn in-depth relations between concepts explained in the lecture and additional solution strategies.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	None	Subject theoretical practical work	andDie Aufgabe wird im Rahmen von Vorlesung und Prüfung definiert. Die Lösung der Aufgabe ist Zulassungsvoraussetzung für die Prüfung.
Examination	Oral exam			
Examination duration and scale	30 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Elective Compulsory			

Course L1979: Model Checking - Proof Engines and Algorithms	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Görschwin Fey
Language	EN
Cycle	SoSe
Content	Correctness is a major concern in embedded systems. Model checking can fully automatically proof formal properties about digital hardware or software. Such properties are given in temporal logic, e.g., to prove "No two orthogonal traffic lights will ever be green." And how do the underlying reasoning algorithms work so effectively in practice despite a computational complexity of NP hardness and beyond? But what are the limitations of model checking? How are the models generated from a given design? The lecture will answer these questions. Open source tools will be used to gather a practical experience. Among other topics, the lecture will consider the following topics: • Modelling digital Hardware, Software, and Cyber Physical Systems • Data structures, decision procedures and proof engines ◦ Binary Decision Diagrams ◦ And-Inverter-Graphs ◦ Boolean Satisfiability ◦ Satisfiability Modulo Theories • Specification Languages ◦ CTL ◦ LTL ◦ System Verilog Assertions • Algorithms for ◦ Reachability Analysis ◦ Symbolic CTL Checking ◦ Bounded LTL-Model Checking ◦ Optimizations, e.g., induction, abstraction • Quality assurance
Literature	Edmund M. Clarke, Jr., Orna Grumberg, and Doron A. Peled. 1999. <i>Model Checking</i>. MIT Press, Cambridge, MA, USA. A. Biere, A. Biere, M. Heule, H. van Maaren, and T. Walsh. 2009. <i>Handbook of Satisfiability: Volume 185 Frontiers in Artificial Intelligence and Applications</i>. IOS Press, Amsterdam, The Netherlands, The Netherlands. Selected research papers

Course L1980: Model Checking - Proof Engines and Algorithms	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Görschwin Fey
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1301: Software Testing				
Courses				
Title	Typ		Hrs/wk	CP
Software Testing (L1791)	Lecture		2	3
Software Testing (L1792)	Project-/problem-based Learning		2	3
Module Responsible	Prof. Sibylle Schupp			
Admission Requirements	None			
Recommended Previous Knowledge	• Software Engineering • Higher Programming Languages • Object-Oriented Programming • Algorithms and Data Structures • Experience with (Small) Software Projects • Statistics			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> Students explain the different phases of testing, describe fundamental techniques of different types of testing, and paraphrase the basic principles of the corresponding test process. They give examples of software development scenarios and the corresponding test type and technique. They explain algorithms used for particular testing techniques and describe possible advantages and limitations. <i>Skills</i> Students identify the appropriate testing type and technique for a given problem. They adapt and execute respective algorithms to execute a concrete test technique properly. They interpret testing results and execute corresponding steps for proper re-test scenarios. They write and analyze test specifications. They apply bug finding techniques for non-trivial problems. Personal Competence <i>Social Competence</i> Students discuss relevant topics in class. They defend their solutions orally. They communicate in English. <i>Autonomy</i> Students can assess their level of knowledge continuously and adjust it appropriately, based on feedback and on self-guided studies. Within limits, they can : own learning goals. Upon successful completion, students can identify and precisely formulate new problems in academic or applied research in the field of : testing. Within this field, they can conduct independent studies to acquire the necessary competencies and compile their findings in academic reports. T devise plans to arrive at new solutions or assess existing ones			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Subject theoretical and practical work			
Examination duration and scale	Software			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory			

Course L1791: Software Testing	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Sibylle Schupp
Language	EN
Cycle	SoSe
Content	- Fundamentals of software testing - Model-based testing - Test automation - Criteria-based testing
Literature	P. Ammann and J. Offutt, "Introduction to Software Testing", 2nd edition 2016. - Selected research papers

Course L1792: Software Testing	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Sibylle Schupp
Language	EN
Cycle	SoSe
Content	- Fundamentals of software testing - Model-based testing - Test automation - Criteria-based testing
Literature	M. Pezze and M. Young, Software Testing and Analysis, John Wiley 2008. P. Ammann and J. Offutt, "Introduction to Software Testing", 2nd edition 2015.

Module M1427: Algorithmic Game Theory				
Courses				
Title	Typ		Hrs/wk	CP
Algorithmic game theory (L2060)	Lecture		3	4
Algorithmic game theory (L2061)	Recitation Section (large)		2	2
Module Responsible	Prof. Matthias Mnich			
Admission Requirements	None			
Recommended Previous Knowledge	- Mathematics I - Mathematics II - Algorithms and Data Structures			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i> - Students can name the basic concepts in algorithmic game theory and mechanism design. They are able to explain them using appropriate examples. - Students can discuss logical connections between these concepts. They are capable of illustrating these connections with the help of examples. - They know game and mechanism design strategies and can reproduce them. <i>Skills</i> - Students can model strategic interaction systems of agents with the help of the concepts studied in this course. Moreover, they are capable of analyzing their efficiency and equilibria, by applying established methods. - Students are able to discover and verify further logical connections between the concepts studied in the course. - For a given problem, the students can develop and execute a suitable approach, and are able to critically evaluate the results. Personal Competence <i>Social Competence</i> - Students are able to work together in teams. They are capable to use mathematics as a common language. - In doing so, they can communicate new concepts according to the needs of their cooperating partners. Moreover, they can design examples to check and deepen the understanding of their peers. <i>Autonomy</i> - Students are capable of checking their understanding of complex concepts on their own. They can specify open questions precisely and know where to get help in solving them. - Students have developed sufficient persistence to be able to work for longer periods in a goal-oriented manner on hard problems.				
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	20 %	Subject	theoretical and practical work
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory			

Course L2060: Algorithmic game theory	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Matthias Mnich
Language	DE/EN
Cycle	SoSe
Content	Algorithmic game theory is a topic at the intersection of economics and computation. It deals with analyzing the behavior and interactions of strategic agents, who often try to maximize their incentives. The environment in which those agents interact is referred to as a game. We wish to understand if the agents can reach an "equilibrium", or steady state of the game, in which agents have no incentive to deviate from their chosen strategies. The algorithmic part is to design efficient methods to find equilibria in games, and to make recommendations to the agents so that they can quickly reach a state of personal satisfaction. We will also study mechanism design. In mechanism design, we wish to design markets and auctions and give strategic options to agents, so that they have an incentive to act rationally. We also wish to design the markets and auctions so that they are efficient, in the sense that all goods are cleared and agents do not overpay for the goods which they acquire. Topics: • basic equilibrium concepts (Nash equilibria, correlated equilibria, ...) • strategic actions (best-response dynamics, no-regret dynamics, ...) • auction design (revenue-maximizing auctions, Vickrey auctions) • stable matching theory (preference aggregations, kidney exchanges, ...) • price of anarchy and selfish routing (Braess' paradox, congestion games, ...)
Literature	• T. Roughgarden: Twenty Lectures on Algorithmic Game Theory, Cambridge University Press, 2016. • N. Nisan, T. Roughgarden, E. Tardos, V. Vazirani. Algorithmic Game Theory. Cambridge University Press, 2007.

Course L2061: Algorithmic game theory	
Typ	Recitation Section (large)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Matthias Mnich
Language	DE/EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1248: Compilers for Embedded Systems				
Courses				
Title	Typ		Hrs/wk	CP
Compilers for Embedded Systems (L1692)	Lecture		3	4
Compilers for Embedded Systems (L1693)	Project-/problem-based Learning		1	2
Module Responsible	Prof. Heiko Falk			
Admission Requirements	None			
Recommended Previous Knowledge	Module "Embedded Systems" C/C++ Programming skills			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	The relevance of embedded systems increases from year to year. Within such systems, the amount of software to be executed on embedded processors grows continuously due to its lower costs and higher flexibility. Because of the particular application areas of embedded systems, highly optimized and application-specific processors are deployed. Such highly specialized processors impose high demands on compilers which have to generate code of highest quality. After the successful attendance of this course, the students are able - to illustrate the structure and organization of such compilers, - to distinguish and explain intermediate representations of various abstraction levels, and - to assess optimizations and their underlying problems in all compiler phases. The high demands on compilers for embedded systems make effective code optimizations mandatory. The students learn in particular, - which kinds of optimizations are applicable at the source code level, - how the translation from source code to assembly code is performed, - which kinds of optimizations are applicable at the assembly code level, - how register allocation is performed, and - how memory hierarchies can be exploited effectively. Since compilers for embedded systems often have to optimize for multiple objectives (e.g., average- or worst-case execution time, energy dissipation, code size), the students learn to evaluate the influence of optimizations on these different criteria. <i>Skills</i> After successful completion of the course, students shall be able to translate high-level program code into machine code. They will be enabled to assess which kind of code optimization should be applied most effectively at which abstraction level (e.g., source or assembly code) within a compiler. While attending the labs, the students will learn to implement a fully functional compiler including optimizations.			
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>				
<i>Students are able to solve similar problems alone or in a group and to present the results accordingly.</i>				
<i>Students are able to acquire new knowledge from specific literature and to associate this knowledge with other classes.</i>				
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Oral exam			
Examination duration and scale	30 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Aircraft Systems Engineering: Core Qualification: Elective Compulsory Aeronautics: Core Qualification: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L1692: Compilers for Embedded Systems	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Heiko Falk
Language	EN
Cycle	SoSe
Content	• Introduction and Motivation • Compilers for Embedded Systems - Requirements and Dependencies • Internal Structure of Compilers • Pre-Pass Optimizations • HIR Optimizations and Transformations • Code Generation • LIR Optimizations and Transformations • Register Allocation • WCET-Aware Compilation • Outlook
Literature	• Peter Marwedel. Embedded System Design - Embedded Systems Foundations of Cyber-Physical Systems. 2nd Edition, Springer, 2012. • Steven S. Muchnick. Advanced Compiler Design and Implementation. Morgan Kaufmann, 1997. • Andrew W. Appel. Modern compiler implementation in C. Oxford University Press, 1998.

Course L1693: Compilers for Embedded Systems	
Typ	Project-/problem-based Learning
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Heiko Falk
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1685: Selected Aspects in Computer Science			
Courses			
Title		Typ	Hrs/wk CP
Selected Aspects in Computer Science (L2672)		Lecture	3 4
Selected Aspects in Computer Science (L2673)		Recitation Section (small)	1 2
Module Responsible	Prof. Görschwin Fey		
Admission Requirements	None		
Recommended Previous Knowledge			
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i> <i>Skills</i> Personal Competence <i>Social Competence</i> <i>Autonomy</i>			
Workload in Hours	Depends on choice of courses		
Credit points	6		
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Elective Compulsory		

Course L2672: Selected Aspects in Computer Science	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Examination Form	Klausur
Examination duration and scale	90 min
Lecturer	Dozenten des SD E
Language	DE/EN
Cycle	WiSe/SoSe
Content	
Literature	

Course L2673: Selected Aspects in Computer Science	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Examination Form	Fachtheoretisch-fachpraktische Arbeit
Examination duration and scale	x
Lecturer	Dozenten des SD E
Language	DE/EN
Cycle	WiSe/SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1682: Secure Software Engineering				
Courses				
Title	Typ		Hrs/wk	CP
Secure Software Engineering (L2667)	Lecture		2	3
Secure Software Engineering (L2668)	Project-/problem-based Learning		2	3
Module Responsible	Prof. Riccardo Scandariato			
Admission Requirements	None			
Recommended Previous Knowledge	Familiarity with basic software engineering concepts (e.g., requirements, design) and basic security concepts (e.g., confidentiality, integrity, availability)			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	Students can: - Elicit security requirements in a software project - Model and document security measures in a software design - Use threat and risk analysis techniques - Understand how security code reviews are performed - Understand the core definitions of concepts related to privacy - Understand privacy enhancing technologies Select appropriate security assurance techniques to be used in a security assurance program			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>	None Students can apply the knowledge acquired throughout the course to the resolution of industrial case studies. Students should also be capable to acquire new knowledge independently from academic publications, technical standards, and white papers.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	5 %	Subject theoretical and practical work	and Gruppenarbeit mit aktuellen Technologien aus dem Bereich Sicherheit
Examination	Written exam			
Examination duration and scale	120 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory			

Course L2667: Secure Software Engineering	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Riccardo Scandariato
Language	EN
Cycle	SoSe
Content	- Secure software development processes and maturity models - Techniques to define security requirements - Techniques to create, document and analyse the design of secure applications - Threat and risk analysis techniques - Security code reviews - Program repair techniques for security vulnerabilities - Privacy engineering
Literature	Charles Haley, Robin Laney, Jonathan Moffett, Bashar Nuseibeh, 2008, Security Requirements Engineering: A Framework for Representation and Analysis, Transactions on Software Engineering Sindre, G. and Opdahl, A.L., 2005. Eliciting security requirements with misuse cases. Requirements engineering, 10(1) Fontaine, P.J., Van Lamsweerde, A., Letier, E. and Darimont, R., 2001. Goal-oriented elaboration of security requirements. Joanna C. S. Santos, Katy Tarrit, Mehdi Mirakhorli, 2017, A Catalog of Security Architecture Weaknesses, IEEE International Conference on Software Architecture Workshops (ICSAW) Jürjens, J., 2002, UMLsec: Extending UML for secure systems development, International Conference on The Unified Modeling Language Lund, M.S., Solhaug, B. and Stølen, K., 2011. A guided tour of the CORAS method. In Model-Driven Risk Analysis, Springer Pratyusa Manadhata, Jeannette Wing, 2021, An Attack Surface Metric, IEEE Transactions on Software Engineering, 37(3) Howard, M.A., 2006. A process for performing security code reviews. IEEE Security & privacy, 4(4) Jaap-Henk Hoepman, 2022, Privacy Design Strategies (The Little Blue Book) Diaz, C. and Gürses, S., 2012. Understanding the landscape of privacy technologies. Proceedings of the information security summit, 12

Course L2668: Secure Software Engineering	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Riccardo Scandariato
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1774: Advanced Internet Computing			
Courses			
Title	Typ	Hrs/wk	CP
Advanced Internet Computing (L2916)	Lecture	2	3
Advanced Internet Computing (L2917)	Project-/problem-based Learning	2	3
Module Responsible	Prof. Stefan Schulte		
Admission Requirements	None		
Recommended Previous Knowledge	Good programming skills are necessary. Previous knowledge in the field of distributed systems is helpful.		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	After successful completion of the course, students are able to: - Describe basic concepts of cloud computing and blockchain technologies - Discuss and assess critical aspects of cloud computing and blockchain technologies - Select and apply cloud and blockchain technologies for particular application areas - Design and develop practical solutions based on cloud and blockchain software The students acquire the ability to model Internet-based distributed systems and to work with these systems. This comprises especially the ability to select and utilize fitting technologies for different application areas. Furthermore, students are able to critically assess the chosen technologies.		
Knowledge			
Skills			
Personal Competence			
Social Competence			
Social Competence	Students can work on complex problems both independently and in teams. They can exchange ideas with each other and use their individual strengths to solve the problem.		
Autonomy	Students are able to independently investigate a complex problem and assess which competencies are required to solve it.		
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Subject theoretical and practical work		
Examination duration and scale	Group project incl. presentation (50 %), written exam (60 min, 50 %)		
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Networks: Elective Compulsory		

Course L2916: Advanced Internet Computing	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Stefan Schulte
Language	EN
Cycle	SoSe
Content	This lecture discusses modern Internet-based distributed systems in two blocks: (i) cloud computing and (ii) blockchain technologies. The following topics will be covered in the single lectures: - Cloud Service and Deployment Models - Virtualization - Further Aspects of Cloud Computing - Edge and Fog Computing - Blockchain technologies - Consensus
Literature	Lecture notes as well as current literature announced in the lecture.

Course L2917: Advanced Internet Computing	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Stefan Schulte
Language	EN
Cycle	SoSe
Content	This project-/problem-oriented part of the module augments the theoretical content of the lecture by a concrete technical problem, which needs to be solved by the students in group work during the semester. Possible topics are (blockchain-based) sensor data integration and blockchain-based voting.
Literature	Lecture notes as well as current literature announced in the lecture.

Module M1773: Cybersecurity Data Science				
Courses				
Title		Typ	Hrs/wk	CP
Cybersecurity Data Science (L2914)		Lecture	2	3
Cybersecurity Data Science (L2915)		Project-/problem-based Learning	2	3
Module Responsible	Prof. Riccardo Scandariato			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge of probabilities and statistics. Familiarity with object oriented programming.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	Students can: • Apply data science methods to the resolution of complex cybersecurity problems. • Use of data science methods to quantify risks and optimize cybersecurity operations. • Identify strengths and limitations of state-of-the-art methods. • Select the performance indicators of data-oriented cybersecurity solutions. • Understand cybersecurity threats in data science methods. Implement and evaluate data-driven models for the identification, treatment, and mitigation of cybersecurity risks			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>	Students can apply the knowledge acquired throughout the course to the resolution of industrial case studies. Students should also be capable to acquire new knowledge independently from academic publications, techical standards, and white papers.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	5 %	Subject practical work	theoretical andGruppenarbeit mit aktuellen Technologien aus dem Bereich Sicherheit
Examination	Written exam			
Examination duration and scale	120 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Elective Compulsory			

Course L2914: Cybersecurity Data Science				
Typ	Lecture			
Hrs/wk	2			
CP	3			
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28			
Lecturer	Prof. Riccardo Scandariato			
Language	EN			
Cycle	SoSe			
Content	Theoretical Foundations: • Introduction to data science • Supervised and unsupervised learning • Data science methods (e.g., clustering, decision trees, artificial neural networks) • Performance indicators • Other data science methods for cybersecurity (e.g., experiments, surveys) Cybersecurity Applications: • Spam/Phishing detection • Intrusion detection • Vulnerability/malware prediction • Adversarial machine learning			
Literature	Sarker, I.H., Kayes, A.S.M., Badsha, S., Alqahtani, H., Watters, P. and Ng, A., 2020. Cybersecurity data science: an overview from machine learning perspective. Journal of Big data, 7(1) Torres, J.M., Comesaña, C.I. and Garcia-Nieto, P.J., 2019. Machine learning techniques applied to cybersecurity. International Journal of Machine Learning and Cybernetics, 10(10) Arp, D., Quiring, E., Pendlebury, F., Warnecke, A., Pierazzi, F., Wressnegger, C., Cavallaro, L. and Rieck, K., Dos and Don'ts of Machine Learning in Computer Security.			

Course L2915: Cybersecurity Data Science	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Riccardo Scandariato
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M0924: Software for Embedded Systems				
Courses				
Title	Typ		Hrs/wk	CP
Software for Embedded Systems (L1069)	Lecture		2	3
Software for Embedded Systems (L1070)	Project/problem-based Learning		3	3
Module Responsible	Prof. Bernd-Christian Renner			
Admission Requirements	None			
Recommended Previous Knowledge	• Very good knowledge and practical experience in programming in the C language and its compilation process • Basic knowledge in software engineering • Basic understanding of assembly language • Basic knowledge of electrical engineering			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	Knowledge • Students know the contents of the module and can reproduce and link them. • They are able to describe the usage and advantages of event-based programming using interrupts. • Students can explain the basic principles and procedures for creating software for embedded systems. • They know the components and functions of a concrete microcontroller. • The participants can explain requirements of real time systems. • They know at least three scheduling algorithms for real time operating systems including their pros and cons. • They know different memories for embedded systems and their use cases, including advantages and disadvantages. • They understand race conditions, their causes and consequences. • They are familiar with at least two bootloader concepts and can evaluate their suitability in different contexts. • They know mechanisms for optimizing the power consumption of microcontrollers with sleep modes. Skills • Students design and write hardware-oriented software modules in a high-level language (C or similar) for an embedded system based on a specific microcontroller. • They can configure and interact with peripherals (timer, ADC, EEPROM), including bit-level register operations, interrupt-based processing, and program flow. • They implement and use a (preemptive) scheduler for an embedded system. • They learn to write independent, reusable software components. They build a complex software system (library and applications) throughout the semester. • They learn how to model event-based software for embedded systems with UML state diagrams and finite state machines and implement a complex system that they modeled. • They can identify race conditions in unknown software and develop strategies to work around them. Personal Competence Social Competence • Students are able to work goal-oriented in small mixed groups. • They learn and broaden their teamwork abilities. • They learn to define and split tasks within the team. • They practice to present their software, the selection of algorithms and their design choices. Autonomy Students are able • to solve assignments related to this lecture independently with instructional direction. • to design, implement, and test software components for an embedded system independently based on a textual description. • to read and understand data sheets and manuals of electronic components (such as micro-controllers and sensors). • to independently transfer the concepts known from the lecture to the exercises and to select algorithms based on given criteria.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	10 %	Attestation	
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Embedded Systems: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L1069: Software for Embedded Systems	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Bernd-Christian Renner
Language	EN
Cycle	SoSe
Content	• General-Purpose Processors • Programming the Atmel AVR • Interrupts and Race Conditions • C for Embedded Systems • Standard Single Purpose Processors: Peripherals • Finite-State Machines • Memory • Operating Systems for Embedded Systems • Real-Time Embedded Systems • Boot loader and Power Management
Literature	1. Embedded System Design, F. Vahid and T. Givargis, John Wiley 2. Programming Embedded Systems: With C and Gnu Development Tools, M. Barr and A. Massa, O'Reilly 3. C und C++ für Embedded Systems, F. Bollow, M. Homann, K. Köhn, MITP 4. The Art of Designing Embedded Systems, J. Ganssle, Newnes 5. Mikrocomputertechnik mit Controllern der Atmel AVR-RISC-Familie, G. Schmitt, Oldenbourg 6. Making Embedded Systems: Design Patterns for Great Software, E. White, O'Reilly

Course L1070: Software for Embedded Systems	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Bernd-Christian Renner
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1810: Autonomous Cyber-Physical Systems				
Courses				
Title	Typ		Hrs/wk	CP
Autonomous Cyber-Physical Systems (L3000)	Lecture		2	3
Autonomous Cyber-Physical Systems (L3001)	Project/problem-based Learning		3	3
Module Responsible	Prof. Bernd-Christian Renner			
Admission Requirements	None			
Recommended Previous Knowledge	• Very good knowledge and practical experience in programming in the C/C++ language (e.g., module: Procedural Programming for Computer Scientists) • Basic knowledge in software engineering • Basic knowledge in wired and wireless communication protocols • Principal understanding of simple electronic circuits			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i>	This master's course centers on the advanced Internet of Things (IoT) systems, spotlighting energy sustainability and Real-Time Operating Systems (RTOS) within cyber-physical frameworks. Key focus areas include the exploration of IoT system architecture, especially leveraging RTOS for real-time responsiveness, and energy-efficient technologies for sustainable operations. Upon completion, students will be capable of: • Outlining the structure and operational principles of IoT systems and their significance in modern cyber-physical environments. • Differentiating between RTOS and conventional operating systems in the context of IoT applications. • Evaluating energy harvesting (specifically solar technologies) and storage systems for their applicability and efficiency in IoT devices • Knowing different wireless technologies (e.g., Bluetooth LE, LoRaWAN, Zigbee) and networking protocols (TCP/IP, MQTT) for IoT connectivity and functionality, and pinpointing their dis-/advantages.			
<i>Skills</i>	Students will acquire practical skills to: • Describe, plan, develop and optimize IoT applications using RTOS, focusing on multitasking, efficient memory, and power management. • Select and deploy suitable wireless and networking technologies considering device requirements and operational contexts. • Integrate energy harvesting and storage technologies into IoT solutions, enhancing their sustainability and autonomy. • Conduct comprehensive planning and evaluation of IoT projects, incorporating scientific methodologies for real-world applications.			
Personal Competence <i>Social Competence</i>	• Students are able to work goal-oriented in small mixed groups. • They learn and broaden their teamwork abilities. • They learn to define and split tasks within the team. • After completing the module, the students are able to work on similar tasks alone or in a group and to present the results in a suitable way.			
<i>Autonomy</i>	After completing the module, the students are able to independently work on sub-areas of the subject using specialist literature, to summarize and present the knowledge they have acquired and to link it to the content of other courses.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	10 %	Attestation	
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Wireless and Sensor Technologies: Elective Compulsory Electrical Engineering: Specialisation Wireless and Sensor Technologies: Elective Compulsory Computer Science in Engineering: Specialisation II. Engineering Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L3000: Autonomous Cyber-Physical Systems	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Bernd-Christian Renner
Language	EN
Cycle	SoSe
Content	
Literature	

Course L3001: Autonomous Cyber-Physical Systems	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Bernd-Christian Renner
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1772: Smart Sensors				
Courses				
Title	Typ		Hrs/wk	CP
Smart Sensors (L2904)	Lecture		2	2
Smart Sensors Lab (L2905)	Project-/problem-based Learning		3	4
Module Responsible	Prof. Ulf Kulau			
Admission Requirements	None			
Recommended Previous Knowledge	• Embedded Systems (recommended) • Computer engineering (recommended) • Programming skills in C (recommended) • FPGA experience (desirable)			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> The term smart sensors is actually easy to explain. While data processing in classic sensor systems usually takes place in the backend, the aim of intelligent sensors is to integrate signal processing ever closer to the actual sensor. However, sensors are resource-limited systems and therefore pose many challenges, which are covered in this course. There is a strong focus on the use of ULP FPGAs as flexible processing elements for smart sensors and how these can be implemented using the Verilog hardware description language. The following contents are taught: • Introduction and motivation • Sensors, uncertainties and their compensation • Analog and digital sensor interfaces • Processing elements for smart sensors • General introduction with a focus on ULP FPGAs • Introduction to HDL Verilog • Signal processing (filters and fusion) • Decision making on smart sensors • Reliability, fault tolerance and fault detection In addition to the lecture, there is an exercise in which a simple sensor system is to be implemented using a ULP FPGA. The knowledge learned in the lecture and especially the excursus on the introduction to HDL Verilog will help. <i>Skills</i> After completing this module, students will have a deeper understanding of how smart sensor systems are constructed, which individual components play a role and how they can design their own components: • They have a deeper understanding of the physical and electrotechnical principles of intelligent (smart) sensors • They know the functionalities of the processing elements that can be used for smart sensors, in particular FPGA • They have received a basic introduction to HDL Verilog and are able to carry out small implementations of a smart sensor • They will be able to plan, model, implement and evaluate sensor-related signal processing • They have acquired basic knowledge for obtaining reliable smart sensors Personal Competence <i>Social Competence</i> As part of the module, concepts learned in small groups are to be implemented on an LP FPGA platform. Students learn to work in a team and develop solutions together. Specific tasks are worked on within the group, with cross-group collaboration (exchange) also taking place. The aim is to implement their own small smart sensor system based on the content of the lecture. The results of each group are presented to each other at the end of the semester. <i>Autonomy</i> After completing this module, students will be able to independently develop, optimize and evaluate solutions for Smart Sensor-Systems based on the knowledge they have acquired and further specialist literature.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	None			
Examination	Oral exam			
Examination duration and scale	25 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Wireless and Sensor Technologies: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Wireless and Sensor Technologies: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Embedded Systems: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L2904: Smart Sensors	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Ulf Kulau
Language	EN
Cycle	SoSe
Content	The following contents are taught: • Introduction and motivation • Sensors, uncertainties and their compensation • Analog and digital sensor interfaces • Processing elements for smart sensors • General introduction with a focus on ULP FPGAs • Introduction to HDL Verilog • Signal processing (filters and fusion) • Decision making on smart sensors • Reliability, fault tolerance and fault detection
Literature	• Czichos, Horst. Mechatronik: Grundlagen und Anwendungen technischer Systeme. Springer-Verlag, 2015. • Czichos, Horst, and Werner Daum. "Messtechnik und Sensorik." Dubbel. Springer, Berlin, Heidelberg, 2007. W1-W37. • Iso, I., and BIPM OIML. "Guide to the Expression of Uncertainty in Measurement." Geneva, Switzerland 122 (1995): 16-17 • Hoffmann, Jörg, and Fachbuchverlag Leipzig im Carl-Hanser-Verlag. Taschenbuch der Messtechnik. München, 2010. • Niebuhr, Johannes, and Gerhard Lindner. "Physikalische Meßtechnik mit Sensoren." tm-Technisches Messen 63 (1996):4. • Kiencke, Uwe, and Ralf Eger. Messtechnik. Springer Berlin Heidelberg, 2008. (https://katalog.tub.tuhh.de/Record/1646558138) • Fraden, Jacob. "Handbook of modern sensors: physics, designs, and applications." (2007). • Meijer, Gerard, Kofi Makinwa, and Michiel Pertijs. Smart sensor systems: Emerging technologies and applications. John Wiley & Sons, 2014. • Vansteenkiste, Elias. New FPGA design tools and architectures. Diss. Ghent University, 2016. • Wüst, Klaus. Mikroprozessortechnik: Grundlagen, Architekturen, Schaltungstechnik und Betrieb von Mikroprozessoren und Mikrocontrollern. Springer-Verlag, 2010. • Huddleston, Creed. Intelligent sensor design using the microchip dsPIC. Elsevier, 2006. • Wolf, Marilyn. Computers as components: principles of embedded computing system design. Elsevier, 2012. • Smith, Douglas J. HDL Chip Design: A practical guide for designing, synthesizing and simulating ASICs and FPGAs • Thomas, Donald, and Philip Moorby. The Verilog® hardware description language. Springer Science & Business Media, 2008. • LaMeres, Brock J. Quick Start Guide to Verilog. Springer International Publishing, 2019. (https://katalog.tub.tuhh.de/Record/1067372644) • J. Hennessy, D. Patterson: „Computer Architecture: A Quantitative Approach“, Morgan Kaufmann, 4. Aufl., 2006, • B. Parhami : „Computer Arithmetic: Algorithms and Hardware Designs“, Oxford Univ Press, 2009 • Pei, Yan, et al. "An elementary introduction to Kalman filtering." Communications of the ACM 62.11 (2019): 122-133 • Johnson, Barry W. "An introduction to the design and analysis of fault-tolerant systems." Fault-tolerant computer system design 1 (1996)

Course L2905: Smart Sensors Lab	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Ulf Kulau
Language	EN
Cycle	SoSe
Content	The following contents are taught in the PBL, drawing on previous knowledge from the lecture. The basic aim of this Smart Sensors Lab is to implement a simple smart sensor system using a ULP FPGA. • Getting to know a ULP FPGA platform (simple introductory tasks) • Implementation of a host interface (UART) • Implementation of a sensor interface SPI • Sensor-near data processing (creative part)
Literature	• Thomas, Donald, and Philip Moorby. The Verilog® hardware description language. Springer Science & Business Media, 2008. • LaMeres, Brock J. Quick Start Guide to Verilog. Springer International Publishing, 2019. (https://katalog.tub.tuhh.de/Record/1067372644) • Online-Simulators: ◦ https://www.jdoodle.com/execute-verilog-online/ ◦ https://www.edaplayground.com/home ◦ https://www.tutorialspoint.com/compile_verilog_online.php

Module M1794: Applied Cryptography				
Courses				
Title	Typ		Hrs/wk	CP
Applied Cryptography (L2954)	Integrated Lecture		3	4
Applied Cryptography (L2955)	Recitation Section (small)		1	2
Module Responsible	Prof. Sibylle Fröschle			
Admission Requirements	None			
Recommended Previous Knowledge	Basics in IT security including cryptography, basics in theoretical computer science including computability and logic			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	The students know and can explain - the core cryptographic primitives and constructions in the rigorous treatment of modern cryptography - cryptographic protocols for authentication and key establishment in the modern rigorous treatment - advanced key management techniques using trust anchors such as hardware security modules - the impact of quantum computers on cryptography and the advance towards post-quantum cryptography			
<i>Skills</i>	The students are able to - understand subtle attacks on inadequately applied cryptography and how to avoid them - follow security standards to develop secure cryptographic applications - systematically analyze and solve complex problems in the area of cryptographic protocols and key management - recognize novel challenges and limitations, e.g. posed by novel types of attacks			
Personal Competence				
<i>Social Competence</i>	The students are able to - expertly discuss when cryptographic solutions are necessary and what it takes to deploy them with experts and non-experts - champion the approach of modern cryptography with its focus on precise definitions and assumptions as essential for security			
<i>Autonomy</i>	The students are able to - follow up and critically assess current developments in cryptography and cryptographic applications including the challenge of post-quantum cryptography - master a new topic within the area by self-study and self-initiated interaction with experts and peers			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	10 %	Exercises	Die Übungsaufgaben finden semesterbegleitend statt
Examination	Written exam			
Examination duration and scale	120 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Networks: Elective Compulsory			

Course L2954: Applied Cryptography	
Typ	Integrated Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Sibylle Fröschle
Language	EN
Cycle	SoSe
Content	Cryptography plays a key role in securing the digital world we live in today. This course provides you with a deep understanding of symmetric and public-key cryptography based on the modern approach to cryptography. The modern approach requires that cryptographic constructions are provably secure relative to precise definitions and plausible assumptions. The assumptions typically stem from computability theory and algorithmic number theory. Moreover, you will get to know how cryptography can be applied to solve security problems such as secure communication over an open channel such as the Internet, and we will cover advanced aspects of practical deployment such as key management anchored in hardware security modules. We will see how the theme of precise definition and provable security carries forth in these areas, and are a must for today's security applications. Finally, we discuss the impact of the advent of quantum computers on cryptography and its application, and the current state of post-quantum cryptography.
Literature	Introduction to Modern Cryptography, Third Edition, Jonathan Katz and Jehuda Lindell, Chapman & Hall/CRC, 2021

Course L2955: Applied Cryptography	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Sibylle Fröschle
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M0839: Traffic Engineering				
Courses				
Title		Typ	Hrs/wk	CP
Network Optimization of Communication Networks (L0902)		Project-/problem-based Learning	2	3
Queueing Theory for Communication Networks (L0900)		Lecture	1	2
Queueing Theory for Communication Networks (L0901)		Recitation Section (small)	1	1
Module Responsible	Prof. Andreas Timm-Giel			
Admission Requirements	None			
Recommended Previous Knowledge	- Fundamentals of communication or computer networks - Stochastics			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> Students are able to describe methods for planning, optimisation and performance evaluation of communication networks. <i>Skills</i> Students are able to solve typical planning and optimisation tasks for communication networks. Furthermore they are able to evaluate the network performance using queueing theory. Students are able to apply independently what they have learned to other and new problems. They can present their results in front of experts and discuss them. Personal Competence <i>Social Competence</i> <i>Autonomy</i> Students are able to acquire the necessary expert knowledge to understand the functionality and performance of new communication networks independently.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Oral exam			
Examination duration and scale	30 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Networks: Elective Compulsory			

Course L0902: Network Optimization of Communication Networks	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Andreas Timm-Giel, Konrad Fuger
Language	EN
Cycle	WiSe
Content	Selected applications of methods for planning, optimization, and performance evaluation of communication networks, which have been introduced in the traffic engineering lecture are prepared by the students and presented in a seminar.
Literature	U. Killat, Entwurf und Analyse von Kommunikationsnetzen, Vieweg + Teubner - further literature announced in the lecture

Course L0900: Queuing Theory for Communication Networks	
Typ	Lecture
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Andreas Timm-Giel, Konrad Fuger
Language	EN
Cycle	WiSe
Content	Network Planning and Optimization • Linear Programming (LP) • Network planning with LP solvers • Planning of communication networks Queueing Theory for Communication Networks • Stochastic processes • Queueing systems • Switches (circuit- and packet switching) • Network of queues
Literature	Literatur: U. Killat, Entwurf und Analyse von Kommunikationsnetzen, Springer Weitere Literatur wird in der Lehrveranstaltung bekanntgegeben / Literature: U. Killat, Entwurf und Analyse von Kommunikationsnetzen, Springer further literature announced in the lecture

Course L0901: Queuing Theory for Communication Networks	
Typ	Recitation Section (small)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Andreas Timm-Giel, Konrad Fuger
Language	EN
Cycle	WiSe
Content	Accompanying exercise for the traffic engineering course
Literature	Literatur: U. Killat, Entwurf und Analyse von Kommunikationsnetzen, Springer Weitere Literatur wird in der Lehrveranstaltung bekanntgegeben / Literature: U. Killat, Entwurf und Analyse von Kommunikationsnetzen, Springer further literature announced in the lecture

Module M2051: Secure Software Application Project				
Courses				
Title	Typ		Hrs/wk	CP
Secure Software Application Project (L3307)	Project-/problem-based Learning		4	6
Module Responsible	Prof. Riccardo Scandariato			
Admission Requirements	None			
Recommended Previous Knowledge	Some familiarity with web programming (e.g., Javascript, PHP). Prior attendance to the courses "Software Security" or "Secure Software Engineering" is beneficial but not mandatory.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i>	Upon the completion of this modules, students will be able to: - Recognize coding pitfalls that have security implications. - Select suitable methods for security assessment. - Interact and collaborate effectively in an agile team.			
<i>Skills</i>	Upon the completion of this modules, students will be able to: - Conduct security evaluations of software systems. - Write secure code. - Fix vulnerable code.			
Personal Competence <i>Social Competence</i>	The project is done in groups and in close cooperation within the teammates is required. Upon the completion of this modules, students will be able to: - Interact and collaborate effectively in an agile team. - Coordinate development effort. - Resolve conflicts and provide constructive feedback			
<i>Autonomy</i>	After completing the module, students will be able to independently carry out secure development software projects.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written elaboration			
Examination duration and scale	Report and presentation about the achieved results			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory			

Course L3307: Secure Software Application Project	
Typ	Project-/problem-based Learning
Hrs/wk	4
CP	6
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Lecturer	Prof. Riccardo Scandariato
Language	EN
Cycle	WiSe/SoSe
Content	This is a hands-on project course (PBL) where students will work together in development teams (e.g. 5 students). The overall goal of the project is to: - Acquire better software development skills. - Adopt secure programming practices in a realistic context. - Incorporate security best practices in software development. A functional prototype of a realistic software application is provided (e.g., a web store written in Python and Javascript) and the teams will have to progress through some development phases, including: (i) fixing functional bugs and security vulnerabilities, (ii) extend the functionality and the security features of the system, and (iii) validate the system via state-of-the-art security assessment techniques (e.g., static security application testing, or threat analysis). Prior attendance to the courses "Software Security" or "Secure Software Engineering" is beneficial but not mandatory. Some introductory (theoretical) material will be provided in this course.
Literature	- Greg Hoglund, Gary McGraw, Exploiting Software: How to Break Code, Addison-Wesley, 2004 - Mark G. Graff, Kenneth R. Van Wyk, Debby Russell, Secure Coding: Principles and Practices, O'Reilly, 2003

Module M2138: Internet of Things				
Courses				
Title		Typ	Hrs/wk	CP
Internet of Things (L3426)		Lecture	2	3
Internet of Things (L3427)		Recitation Section (small)	2	3
Module Responsible	Prof. Olaf Landsiedel			
Admission Requirements	None			
Recommended Previous Knowledge				
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	Knowledge - Understanding of the foundational concepts and challenges of wireless networking and the Internet of Things (IoT). - Knowledge of low-power wireless communication technologies, such as NFC, BLE, and 802.15.4. - Understanding of IoT networking protocols like RPL, TSCH, and Thread. - Knowledge of operating systems for IoT and security aspects in IoT. - Understanding of IoT applications and the mechanisms they use to provide services. - Ability to analyze the challenges and requirements of different approaches in the IoT field. - Ability to compare and summarize the strengths and weaknesses of individual IoT mechanisms. Skills - Development and evaluation of small-scale wireless networks and IoT systems & applications. - Practical application of fundamental mechanisms introduced in the lectures. - Demonstration of software developments considering unreliable wireless links and resource constraints (e.g., battery-driven). Personal Competence Social Competence teamwork and collaborative project work through the "hands-on experience in developing IoT systems & applications." Autonomy - Ability to independently analyze and evaluate existing and new methods for IoT system design. - Application of learned concepts to real-world problems. - Independently developing solutions for challenges in the design and implementation of IoT systems, especially in terms of resource-constrained and low-power networks.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	None	Attestation	
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory			

Course L3426: Internet of Things	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Olaf Landsiedel
Language	EN
Cycle	WiSe
Content	Abstract: This course will introduce and discuss the underlying concepts and mechanisms that drive wireless networks and the Internet of Things (IoT). The lectures provide you with the required knowledge, and the labs give you a hands-on experience in developing networking and applications in the Internet of Things and exploring their real-world challenges. Objectives: 1. Knowledge and understanding: Show knowledge of basic concepts and challenges of wireless networking and the Internet of Things, low-power wireless communication, NFC, BLE and 802.15.4), IoT networking (RPL, TSCH, and Thread), Operating Systems for IoT, Security in IoT, and IoT Applications. Describe applications of the Internet of Things and the mechanisms these use to provide their services. Discuss and analyze the challenges and requirements that the different approaches have. Compare and summarize the strength and weaknesses associated with the individual mechanisms. 2. Skills and abilities: Develop and evaluate small-scale wireless networks and IoT systems & applications using fundamental mechanisms introduced in the lectures. Demonstration of these software developments in advanced settings including unreliable wireless links and resource constraints (e.g., battery driven). 3. Judgment and approach: Describe and analyze existing and new methods for IoT systems and application design. In particular, the system's ability for low-power wireless networking and operation under strong resource constraints. Contents: Aim: What is the Internet of Things and its applications? How can we build reliable and resource efficient IoT systems and applications? How do its devices in the Internet of Things communicate? How do wireless networks work? These are a few questions that this course addresses. The goal of the courses is to understand the design of the Internet of Things and discuss the underlying principles and mechanisms. Our lectures provide you with the required fundamentals, and our labs give you a hands-on experience in developing IoT systems and applications and exploring their real-world challenges. Content: We begin the course with an introduction to basic concepts of IoT Systems and Applications and the challenges they pose. We continue with the main course content and focus on: • Wireless networking • low-power wireless communication (NFC, BLE and 802.15.4) • IoT networking (RPL, TSCH, and Thread) • web-technologies for IoT such as COAP • Operating Systems for IoT • Security in IoT • Selected IoT Applications Our lectures provide students with the required fundamentals, and exercise and projects give students a hands-on experience in developing IoT systems & applications and exploring their real-world challenges. This course offers learning experiences that involve hands-on experimentation and analysis as they reinforce student understanding of concepts and their application to real-world problems. Overall, this course provides the students the ability to understand fundamental issues in the design of methods for IoT systems and applications.
Literature	

Course L3427: Internet of Things	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Olaf Landsiedel
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Specialization II: Intelligence Engineering

Module M0633: Industrial Process Automation

Courses

Title	Typ	Hrs/wk	CP
Industrial Process Automation (L0344)	Lecture	2	3
Industrial Process Automation (L0345)	Recitation Section (small)	2	3

Module Responsible	Prof. Alexander Schläefer
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Admission Requirements	None
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Recommended Previous Knowledge	mathematics and optimization methods principles of automata principles of algorithms and data structures programming skills
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Educational Objectives	After taking part successfully, students have reached the following learning results
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Professional Competence	
<i>Knowledge</i>	The students can evaluate and assess discrete event systems. They can evaluate properties of processes and explain methods for process analysis. The students can compare methods for process modelling and select an appropriate method for actual problems. They can discuss scheduling methods in the context of actual problems and give a detailed explanation of advantages and disadvantages of different programming methods. The students can relate process automation to methods from robotics and sensor systems as well as to recent topics like 'cyberphysical systems' and 'industry 4.0'.
<i>Skills</i>	The students are able to develop and model processes and evaluate them accordingly. This involves taking into account optimal scheduling, understanding algorithmic complexity, and implementation using PLCs.
Personal Competence	
<i>Social Competence</i>	The students can independently define work processes within their groups, distribute tasks within the group and develop solutions collaboratively.
<i>Autonomy</i>	The students are able to assess their level of knowledge and to document their work results adequately.

Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
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Credit points	6
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Course achievement	Compulsory	Bonus	Form	Description
	No	10 %	Exercises	

Examination	Written exam
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Examination duration and scale	90 minutes
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Assignment for the Following Curricula	Bioprocess Engineering: Specialisation A - General Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation General Process Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical Process Engineering: Elective Compulsory Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Elective Compulsory Electrical Engineering: Specialisation Control and Power Systems Engineering: Elective Compulsory Aircraft Systems Engineering: Core Qualification: Elective Compulsory International Management and Engineering: Specialisation II. Mechatronics: Elective Compulsory International Management and Engineering: Specialisation II. Product Development and Production: Elective Compulsory Mechanical Engineering and Management: Specialisation Mechatronics: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory Process Engineering: Specialisation Chemical Process Engineering: Elective Compulsory Process Engineering: Specialisation Process Engineering: Elective Compulsory
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Course L0344: Industrial Process Automation	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Alexander Schlaefer
Language	EN
Cycle	WiSe
Content	- foundations of problem solving and system modeling, discrete event systems - properties of processes, modeling using automata and Petri-nets - design considerations for processes (mutex, deadlock avoidance, liveness) - optimal scheduling for processes - optimal decisions when planning manufacturing systems, decisions under uncertainty - software design and software architectures for automation, PLCs
Literature	J. Lunze: „Automatisierungstechnik“, Oldenbourg Verlag, 2012 Reisig: Petrinetze: Modellierungstechnik, Analysemethoden, Fallstudien; Vieweg+Teubner 2010 Hrúz, Zhou: Modeling and Control of Discrete-event Dynamic Systems; Springer 2007 Li, Zhou: Deadlock Resolution in Automated Manufacturing Systems, Springer 2009 Pinedo: Planning and Scheduling in Manufacturing and Services, Springer 2009

Course L0345: Industrial Process Automation	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Alexander Schlaefer
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0629: Intelligent Autonomous Agents and Cognitive Robotics				
Courses				
Title	Typ		Hrs/wk	CP
Intelligent Autonomous Agents and Cognitive Robotics (L0341)	Lecture		2	4
Intelligent Autonomous Agents and Cognitive Robotics (L0512)	Recitation Section (small)		2	2
Module Responsible	Rainer Marrone			
Admission Requirements	None			
Recommended Previous Knowledge	Vectors, matrices, Calculus			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	Students can explain the agent abstraction, define intelligence in terms of rational behavior, and give details about agent design (goals, utilities, environments). They can describe the main features of environments. The notion of adversarial agent cooperation can be discussed in terms of decision problems and algorithms for solving these problems. For dealing with uncertainty in real-world scenarios, students can summarize how Bayesian networks can be employed as a knowledge representation and reasoning formalism in static and dynamic settings. In addition, students can define decision making procedures in simple and sequential settings, with and with complete access to the state of the environment. In this context, students can describe techniques for solving (partially observable) Markov decision problems, and they can recall techniques for measuring the value of information. Students can identify techniques for simultaneous localization and mapping, and can explain planning techniques for achieving desired states. Students can explain coordination problems and decision making in a multi-agent setting in term of different types of equilibria, social choice functions, voting protocol, and mechanism design techniques.			
<i>Skills</i>	Students can select an appropriate agent architecture for concrete agent application scenarios. For simplified agent application students can derive decision trees and apply basic optimization techniques. For those applications they can also create Bayesian networks/dynamic Bayesian networks and apply bayesian reasoning for simple queries. Students can also name and apply different sampling techniques for simplified agent scenarios. For simple and complex decision making students can compute the best action or policies for concrete settings. In multi-agent situations students will apply techniques for finding different equilibria states, e.g., Nash equilibria. For multi-agent decision making students will apply different voting protocols and compare and explain the results.			
Personal Competence				
<i>Social Competence</i>	Students are able to discuss their solutions to problems with others. They communicate in English			
<i>Autonomy</i>	Students are able of checking their understanding of complex concepts by solving variants of concrete problems			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written exam			
Examination duration and scale	90 minutes			
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory Biomedical Engineering: Specialisation Implants and Endoprostheses: Elective Compulsory Biomedical Engineering: Specialisation Medical Technology and Control Theory: Elective Compulsory Biomedical Engineering: Specialisation Management and Business Administration: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L0341: Intelligent Autonomous Agents and Cognitive Robotics	
Typ	Lecture
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Rainer Marrone
Language	EN
Cycle	WiSe
Content	• Definition of agents, rational behavior, goals, utilities, environment types • Adversarial agent cooperation: Agents with complete access to the state(s) of the environment, games, Minimax algorithm, alpha-beta pruning, elements of chance • Uncertainty: Motivation: agents with no direct access to the state(s) of the environment, probabilities, conditional probabilities, product rule, Bayes rule, full joint probability distribution, marginalization, summing out, answering queries, complexity, independence assumptions, naive Bayes, conditional independence assumptions • Bayesian networks: Syntax and semantics of Bayesian networks, answering queries revised (inference by enumeration), typical-case complexity, pragmatics: reasoning from effect (that can be perceived by an agent) to cause (that cannot be directly perceived). • Probabilistic reasoning over time: Environmental state may change even without the agent performing actions, dynamic Bayesian networks, Markov assumption, transition model, sensor model, inference problems: filtering, prediction, smoothing, most-likely explanation, special cases: hidden Markov models, Kalman filters, Exact inferences and approximations • Decision making under uncertainty: Simple decisions: utility theory, multivariate utility functions, dominance, decision networks, value of information Complex decisions: sequential decision problems, value iteration, policy iteration, MDPs Decision-theoretic agents: POMDPs, reduction to multidimensional continuous MDPs, dynamic decision networks • Simultaneous Localization and Mapping • Planning • Game theory (Golden Balls: Split or Share) Decisions with multiple agents, Nash equilibrium, Bayes-Nash equilibrium • Social Choice Voting protocols, preferences, paradoxes, Arrow's Theorem, • Mechanism Design Fundamentals, dominant strategy implementation, Revelation Principle, Gibbard-Satterthwaite Impossibility Theorem, Direct mechanisms, incentive compatibility, strategy-proofness, Vickrey-Groves-Clarke mechanisms, expected externality mechanisms, participation constraints, individual rationality, budget balancedness, bilateral trade, Myerson-Satterthwaite Theorem
Literature	1. Artificial Intelligence: A Modern Approach (Third Edition), Stuart Russell, Peter Norvig, Prentice Hall, 2010, Chapters 2-5, 10-11, 13-17 2. Probabilistic Robotics, Thrun, S., Burgard, W., Fox, D. MIT Press 2005 3. Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations, Yoav Shoham, Kevin Leyton-Brown, Cambridge University Press, 2009

Course L0512: Intelligent Autonomous Agents and Cognitive Robotics	
Typ	Recitation Section (small)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Rainer Marrone
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1598: Image Processing				
Courses				
Title	Typ		Hrs/wk	CP
Image Processing (L2443)	Lecture		2	4
Image Processing (L2444)	Recitation Section (small)		2	2
Module Responsible	Prof. Tobias Knopp			
Admission Requirements	None			
Recommended Previous Knowledge	Signal and Systems			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> The students know about • visual perception • multidimensional signal processing • sampling and sampling theorem • filtering • image enhancement • edge detection • multi-resolution procedures: Gauss and Laplace pyramid, wavelets • image compression • image segmentation • morphological image processing <i>Skills</i> The students can • analyze, process, and improve multidimensional image data • implement simple compression algorithms • design custom filters for specific applications Personal Competence <i>Social Competence</i> Students can work on complex problems both independently and in teams. They can exchange ideas with each other and use their individual strengths to solve the problem. <i>Autonomy</i> Students are able to independently investigate a complex problem and assess which competencies are required to solve it.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Data Science: Specialisation I. Mathematics/Computer Science: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Medical Technology: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Medical Technology: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Signal Processing: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Communication and Signal Processing: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L2443: Image Processing	
Typ	Lecture
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Prof. Tobias Knopp
Language	EN
Cycle	WiSe
Content	• Visual perception • Multidimensional signal processing • Sampling and sampling theorem • Filtering • Image enhancement • Edge detection • Multi-resolution procedures: Gauss and Laplace pyramid, wavelets • Image Compression • Segmentation • Morphological image processing
Literature	Bredies/Lorenz, Mathematische Bildverarbeitung, Vieweg, 2011 Pratt, Digital Image Processing, Wiley, 2001 Bernd Jähne: Digitale Bildverarbeitung - Springer, Berlin 2005

Course L2444: Image Processing	
Typ	Recitation Section (small)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Tobias Knopp
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1725: Scientific Working in Computational Engineering				
Courses				
Title	Scientific Working in Computational Engineering (L2764)		Typ	Hrs/wk
			Project-/problem-based Learning	6
Module Responsible	Prof. Kay Smarsly			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge in scientific writing. String interest in topics related to computing in civil engineering.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	The students will learn to apply concepts and methods of scientific working in computational engineering. In interaction with the course instructors and in collaboration with each other, the students will also learn to understand the complex process of scientific thinking, being able to accurately plan, implement and analyze scientific projects, such as prospective master theses. A project will be conducted throughout the semester, which will contribute to the grade. Since scientific writing is of particular importance in this course, a scientific paper will be developed based, which is a prerequisite for the final examination. The paper will be written based on the project conducted within this course. Project meetings in small groups, presentations, and critical discussions of scientific publications are further key activities.			
<i>Skills</i>	The students will be capable (i) of solving a scientific problem following a scientific methodology, (ii) of documenting their work effectively in the form of a paper, and (iii) of sharing their work in a presentation.			
Personal Competence				
<i>Social Competence</i>	The students will be able to work in a multidisciplinary team and develop communication skills necessary for problem solving.			
<i>Autonomy</i>	The students will be able to extend their knowledge and apply it to solve scientific problems by working independently in a project.			
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84			
Credit points	6			
Course achievement	None			
Examination	Written elaboration			
Examination duration and scale	10 pages of work with 15-minute oral presentation			
Assignment for the Following Curricula	Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory Civil Engineering: Specialisation Coastal Engineering: Elective Compulsory Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Civil Engineering: Specialisation Computational Engineering: Elective Compulsory Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory			

Course L2764: Scientific Working in Computational Engineering	
Typ	Project-/problem-based Learning
Hrs/wk	6
CP	6
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84
Lecturer	Prof. Kay Smarsly
Language	EN
Cycle	WiSe/SoSe
Content	In the course, a scientific problem of practical relevance will first be defined, taking into account the interests of the students participating in the course. The scientific problem will then systematically be solved within the framework of a comprehensive project. The principles of scientific working will be taught based on the scientific problem defined previously. As an integral part of scientific working, fundamentals of scientific writing will be presented and applied to a scientific paper to be written during the course. Topics related to scientific writing include structuring in scientific writing (structuring the abstract, the introduction, the main part, the summary and conclusions, and the acknowledgments and references) and recommendations on effective scientific writing (principles of composition, use of English in scientific writing, useful tips, creating figures, writing in mathematics, referencing, and formal email correspondence). A final paper and a final presentation will be assembled by the students.
Literature	Smarsly, K. & Dragos, K., 2019. Scientific Writing in Engineering. Tredition, Hamburg, Germany.

Module M0630: Robotics and Navigation in Medicine				
Courses				
Title		Type	Hrs/wk	CP
Robotics and Navigation in Medicine (L0335)		Lecture	2	3
Robotics and Navigation in Medicine (L0338)		Project Seminar	2	2
Robotics and Navigation in Medicine (L0336)		Recitation Section (small)	1	1
Module Responsible	Prof. Alexander Schlaefer			
Admission Requirements	None			
Recommended Previous Knowledge	- principles of math (algebra, analysis/calculus) - principles of programming, e.g., in Java or C++ - solid R or Matlab skills			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
Knowledge	The students can explain kinematics and tracking systems in clinical contexts and illustrate systems and their components in detail. Systems can be evaluated with respect to collision detection and safety and regulations. Students can assess typical systems regarding design and limitations.			
Skills	The students are able to design and evaluate navigation systems and robotic systems for medical applications.			
Personal Competence				
Social Competence	The students are able to grasp practical tasks in groups, develop solution strategies independently, define work processes and work on them collaboratively. The students are able to collaboratively organize their work processes and software solutions using virtual communication and software management tools. The students can critically reflect on the results of other groups, make constructive suggestions for improvement, and also incorporate them into their own work.			
Autonomy	The students can assess their level of knowledge and independently control their learning processes on this basis as well as document their work results. They can critically evaluate the results achieved and present them in an appropriate argumentative manner to the other groups.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	10 %	Written elaboration	
	Yes	10 %	Presentation	
Examination	Written exam			
Examination duration and scale	90 minutes			
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Data Science: Specialisation III. Applications: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Medical Technology: Elective Compulsory Electrical Engineering: Specialisation Medical Technology: Elective Compulsory Computer Science in Engineering: Specialisation II. Engineering Science: Elective Compulsory International Management and Engineering: Specialisation II. Electrical Engineering: Elective Compulsory International Management and Engineering: Specialisation II. Process Engineering and Biotechnology: Elective Compulsory International Management and Engineering: Specialisation II. Medical Engineering: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Product Development: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Production: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Materials: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory Biomedical Engineering: Specialisation Implants and Endoprostheses: Elective Compulsory Biomedical Engineering: Specialisation Medical Technology and Control Theory: Elective Compulsory Biomedical Engineering: Specialisation Management and Business Administration: Elective Compulsory Product Development, Materials and Production: Specialisation Product Development: Elective Compulsory Product Development, Materials and Production: Specialisation Production: Elective Compulsory Product Development, Materials and Production: Specialisation Materials: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Bio- and Medical Technology: Elective Compulsory			

Course L0335: Robotics and Navigation in Medicine	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Alexander Schläefer
Language	EN
Cycle	SoSe
Content	- kinematics - calibration - tracking systems - navigation and image guidance - motion compensation The seminar extends and complements the contents of the lecture with respect to recent research results.
Literature	Spong et al.: Robot Modeling and Control, 2005 Troccaz: Medical Robotics, 2012 Further literature will be given in the lecture.

Course L0338: Robotics and Navigation in Medicine	
Typ	Project Seminar
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Alexander Schläefer
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Course L0336: Robotics and Navigation in Medicine	
Typ	Recitation Section (small)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Alexander Schläefer
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M0627: Machine Learning and Data Mining			
Courses			
Title	Typ	Hrs/wk	CP
Machine Learning and Data Mining (L0340)	Lecture	2	4
Machine Learning and Data Mining (L0510)	Recitation Section (small)	2	2
Module Responsible	Rainer Marrone		
Admission Requirements	None		
Recommended Previous Knowledge	• Calculus • Stochastics		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i>	Students can explain the difference between instance-based and model-based learning approaches, and they can enumerate basic machine learning technique for each of the two basic approaches, either on the basis of static data, or on the basis of incrementally incoming data . For dealing with uncertainty, students can describe suitable representation formalisms, and they explain how axioms, features, parameters, or structures used in these formalisms can be learned automatically with different algorithms. Students are also able to sketch different clustering techniques. They depict how the performance of learned classifiers can be improved by ensemble learning, and they can summarize how this influences computational learning theory. Algorithms for reinforcement learning can also be explained by students.		
<i>Skills</i>	Student derive decision trees and, in turn, propositional rule sets from simple and static data tables and are able to name and explain basic optimization techniques. They present and apply the basic idea of first-order inductive learning. Students apply the BME, MAP, ML, and EM algorithms for learning parameters of Bayesian networks and compare the different algorithms. They also know how to carry out Gaussian mixture learning. They can contrast kNN classifiers, neural networks, and support vector machines, and name their basic application areas and algorithmic properties. Students can describe basic clustering techniques and explain the basic components of those techniques. Students compare related machine learning techniques, e.g., k-means clustering and nearest neighbor classification. They can distinguish various ensemble learning techniques and compare the different goals of those techniques.		
Personal Competence <i>Social Competence</i> <i>Autonomy</i>			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	90 minutes		
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory		

Course L0340: Machine Learning and Data Mining	
Typ	Lecture
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Rainer Marrone
Language	EN
Cycle	SoSe
Content	• Decision trees • First-order inductive learning • Incremental learning: Version spaces • Uncertainty • Bayesian networks • Learning parameters of Bayesian networks BME, MAP, ML, EM algorithm • Learning structures of Bayesian networks • Gaussian Mixture Models • kNN classifier, neural network classifier, support vector machine (SVM) classifier • Clustering Distance measures, k-means clustering, nearest neighbor clustering • Kernel Density Estimation • Ensemble Learning • Reinforcement Learning • Computational Learning Theory
Literature	1. Artificial Intelligence: A Modern Approach (Third Edition), Stuart Russel, Peter Norvig, Prentice Hall, 2010, Chapters 13, 14, 18-21 2. Machine Learning: A Probabilistic Perspective, Kevin Murphy, MIT Press 2012

Course L0510: Machine Learning and Data Mining	
Typ	Recitation Section (small)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Rainer Marrone
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1871: Big Data				
Courses				
Title	Typ		Hrs/wk	CP
Big Data (L3101)	Lecture		2	3
Big Data (L3102)	Project-/problem-based Learning		2	3
Module Responsible	Prof. Stefan Schulte			
Admission Requirements	None			
Recommended Previous Knowledge	• Good software engineering and implementation skills are very important for the practical part of this module • Previous knowledge in the fields of Distributed Systems and Databases is helpful			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> After successful completion of the course, students are able to: • Describe basic concepts of methods and technologies to process and store very large amounts of data • Discuss and assess critical aspects of Big Data • Select and apply Big Data technologies for particular application areas • Design and develop practical solutions for the processing of very large amounts of data • Implement Big Data services <i>Skills</i> The students acquire the ability to model Big Data systems and to work with these systems. This comprises especially the ability to select and utilize fitting technologies for different application areas. Furthermore, students are able to critically assess the chosen technologies. Personal Competence <i>Social Competence</i> Students can work on complex problems both independently and in teams. They can exchange ideas with each other and use their individual strengths to solve the problem. <i>Autonomy</i> Students are able to independently investigate a complex problem and assess which competencies are required to solve it.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Subject theoretical and practical work			
Examination duration and scale	Group project incl. presentation, written exam			
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Data Science: Core Qualification: Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Signal Processing: Elective Compulsory			

Course L3101: Big Data	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Stefan Schulte
Language	EN
Cycle	SoSe
Content	This lecture discusses the fundamental concepts of modern Big Data systems. The following topics will be covered in the single lectures: • Cloud Computing • Data models for Big Data • NoSQL databases • Lambda architecture including batch, speed, and serving layers • Kappa architecture • Concepts and tools for data stream processing • Current software tools for the processing of very large amounts of data
Literature	Lecture notes as well as current literature, are announced in the lecture.

Course L3102: Big Data	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Stefan Schulte
Language	EN
Cycle	SoSe
Content	This project-/problem-oriented part of the module augments the theoretical content of the lecture with a concrete technical problem, which needs to be solved by the students in group work during the semester. The topics are from the field of Big Data, and address for instance data stream processing or batch processing.
Literature	Lecture notes as well as current literature announced in the lecture.

Module M1724: Smart Monitoring				
Courses				
Title	Typ		Hrs/wk	CP
Smart Monitoring (L2762)	Integrated Lecture		2	2
Smart Monitoring (L2763)	Recitation Section (small)		2	4
Module Responsible	Prof. Kay Smarsly			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge or interest in object-oriented modeling, programming, and sensor technologies are helpful. Interest in modern research and teaching areas, such as Internet of Things, Industry 4.0 and cyber-physical systems, as well as the will to deepen skills of scientific working, are required. Basic knowledge in scientific writing and good English skills.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	The students will become familiar with the principles and practices of smart monitoring. The students will be able to design decentralized smart systems to be applied for continuous (remote) monitoring of systems in the built and in the natural environment. In addition, the students will learn to design and to implement intelligent sensor systems using state-of-the-art data analysis techniques, modern software design concepts, and embedded computing methodologies. Besides lectures, project work is also part of this module, which will be conducted throughout the semester and will contribute to the grade. In small groups, the students will design smart monitoring systems that integrate a number of "intelligent" sensors to be implemented by the students. Specific focus will be put on the application of machine learning techniques. The smart monitoring systems will be mounted on real-world (built or natural) systems, such as bridges or slopes, or on scaled lab structures for validation purposes. The outcome of every group will be documented in a paper. All students of this module will "automatically" participate with their smart monitoring system in the annual "Smart Monitoring" competition. The written papers and oral examinations form the final grades. The module will be taught in English. Limited enrollment.			
<i>Skills</i>	The students will gain insights into operating state-of-the-art smart sensor systems, used for monitoring a wide range of physical processes relevant to engineering, such as environmental, structural, or comfort monitoring. The students will be capable of devising monitoring strategies of physical processes as part of group projects, tailored to their knowledge backgrounds, and to implement the strategies in smart wireless sensor nodes, using embedded computing and programming. Finally, the students will be able to document the findings of their projects in short reports.			
Personal Competence				
<i>Social Competence</i>	The students will be able to work in groups, share parts of the work for their projects, and develop communication skills, towards achieving the common project goals.			
<i>Autonomy</i>	The students will be able to gain a solid basis on approaching and solving problems in engineering, as well as on documenting results, through their involvement in their monitoring group projects.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written elaboration			
Examination duration and scale	10 pages of work with 15-minute oral presentation			
Assignment for the Following Curricula	Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory Civil Engineering: Specialisation Coastal Engineering: Elective Compulsory Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Environmental Engineering: Specialisation Energy and Resources: Elective Compulsory Environmental Engineering: Specialisation Environment and Climate: Elective Compulsory Environmental Engineering: Specialisation Water Quality and Water Engineering: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory Water and Environmental Engineering: Specialisation Cities: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory			

Course L2762: Smart Monitoring	
Typ	Integrated Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Kay Smarsly
Language	EN
Cycle	SoSe
Content	In this course, principles of smart monitoring will be taught, focusing on modern concepts of data acquisition, data storage, and data analysis. Also, fundamentals of intelligent sensors and embedded computing will be illuminated. Autonomous software and decentralized data processing are further crucial parts of the course, including concepts of the Internet of Things, Industry 4.0 and cyber-physical systems. Furthermore, measuring principles, data acquisition systems, data management and data analysis algorithms will be discussed. Besides the theoretical background, numerous practical examples will be shown to demonstrate how smart monitoring may advantageously be used for assessing the condition of systems in the built or natural environment.
Literature	The course contents couples different fields, such as signal processing, sensing technologies, data analytics, environmental engineering, civil engineering, artificial intelligence, database systems, and many more. The basics will be taught in this course. However, specific literature that covers all these topics does not exist. Instead, literature will be referenced in the lectures, all of which are papers that are freely available online.

Course L2763: Smart Monitoring	
Typ	Recitation Section (small)
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Prof. Kay Smarsly
Language	EN
Cycle	SoSe
Content	The contents of the exercises are based on the lecture contents. In addition to the exercises, project work will be conducted throughout the semester, which will consume the majority of the workload. As part of the project work, students will design smart monitoring systems that will be tested in the laboratory or in the field. As mentioned in the module description, the students will participate in the "Smart Monitoring" competition, hosted annually by the Institute of Digital and Autonomous Construction. Students are encouraged to contribute their own ideas. The tools required to implement the smart monitoring systems will be taught in the group exercises as well as through external sources, such as video tutorials and literature.
Literature	The course contents couples different fields, such as signal processing, sensing technologies, data analytics, environmental engineering, civil engineering, artificial intelligence, database systems, and many more. The basics will be taught in this course. However, specific literature that covers all these topics does not exist. Instead, literature will be referenced in the lectures, all of which are papers that are freely available online.

Module M1666: Intelligent Systems Lab				
Courses				
Title	Typ		Hrs/wk	CP
Intelligent Systems Lab (L2709)	Project-/problem-based Learning		6	6
Module Responsible	Prof. Alexander Schlaefer			
Admission Requirements	None			
Recommended Previous Knowledge	Very good programming skills Good knowledge in mathematics Prior knowledge in machine learning is very helpful Prior knowledge in image processing / computer vision is helpful Prior knowledge in robotics is very helpful Prior knowledge in microprocessor programming is helpful			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	Students will be able to explain aspects of intelligent systems (e.g. autonomy, sensing the environment, interacting with the environment) and provide links to ai / robotics / machine learning / computer vision. Students can analyze a complex application scenario and use artificial intelligence methods (particularly from robotics, machine learning, computer vision) to implement an intelligent system. Furthermore, students will be able to define criteria to assess the function of the system and evaluate the system. The students can define project aims and scope and organize the project as team work. They can present their results in an appropriate manner. The students take responsibility for their tasks and coordinate their individual work with other group members. They deliver their work on time. They independently acquire additional knowledge by doing a specific literature research.			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>				
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	None	Group discussion	
Examination	Written elaboration			
Examination duration and scale	approx. 8 pages, time frame: over the course of the semester			
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Computer Science in Engineering: Specialisation II. Engineering Science: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L2709: Intelligent Systems Lab	
Typ	Project-/problem-based Learning
Hrs/wk	6
CP	6
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84
Lecturer	Prof. Alexander Schlaefer
Language	DE/EN
Cycle	SoSe
Content	The actual project topic will be defined as part of the project.
Literature	Wird in der Veranstaltung bekannt gegeben.

Module M2029: Process Imaging				
Courses				
Title	Typ		Hrs/wk	CP
Process Imaging (L2723)	Lecture		3	3
Applied Process Imaging (L2724)	Project-/problem-based Learning		3	3
Module Responsible	Prof. Alexander Penn			
Admission Requirements	None			
Recommended Previous Knowledge	No special prerequisites needed. An interest in imaging techniques and image processing is helpful but not mandatory.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	The module focuses primarily on discussing established imaging techniques including (a) optical and infrared imaging, (b) magnetic resonance imaging, (c) X-ray imaging and tomography. Moreover, it presents and discusses a range of more recent imaging modalities. The students will learn: 1. what these imaging techniques can measure (such as sample density or concentration, material transport, chemical composition, temperature), 2. how the measurement techniques work (physical measurement principles, hardware requirements, image reconstruction), and 3. how to determine the most suited imaging methods for a given problem. After the successful completion of the course, the students shall: 1. understand the physical principles and practical aspects of the most common imaging methods, 2. be able to assess the pros and cons of these methods with regard to cost, complexity, expected contrasts, spatial and temporal resolution, and based on this assessment 3. be able to identify the most suited imaging modality for any specific engineering challenge in the field of chemical and bioprocess engineering.			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>	In the problem-based interactive course, students work in small teams and set up two process imaging systems and use these systems to measure relevant process parameters in different chemical and bioprocess engineering applications. The teamwork will foster interpersonal communication skills.			
<i>Autonomy</i>	Students are guided to work in self-motivation due to the challenge-based character of this module. A final presentation improves presentation skills.			
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84			
Credit points	6			
Course achievement	None			
Examination	Subject theoretical and practical work			
Examination duration and scale	70% written examination, 30% active participation and final presentation of the problem-based learning units with a 5-10 page report			
Assignment for the Following Curricula	Bioprocess Engineering: Specialisation A - General Bioprocess Engineering: Elective Compulsory Bioprocess Engineering: Specialisation B - Industrial Bioprocess Engineering: Elective Compulsory Bioprocess Engineering: Specialisation C - Bioeconomic Process Engineering, Focus Energy and Bioprocess Technology: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical and Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical and Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Core Qualification: Elective Compulsory Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Signal Processing: Elective Compulsory International Management and Engineering: Specialisation II. Process Engineering and Biotechnology: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory Process Engineering: Specialisation Process Engineering: Elective Compulsory Process Engineering: Specialisation Chemical Process Engineering: Elective Compulsory Process Engineering: Specialisation Environmental Process Engineering: Elective Compulsory			

Course L2723: Process Imaging	
Typ	Lecture
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Alexander Penn
Language	EN
Cycle	SoSe
Content	The lecture focuses primarily on presenting and discussing established imaging techniques relevant to the field of engineering including (a) optical and infrared imaging, (b) magnetic resonance imaging, (c) X-ray imaging and tomography. Moreover, it presents and discusses a range of more recent imaging modalities. The students will learn: 1. what these imaging techniques can measure (such as sample density or concentration, material transport, chemical composition, temperature), 2. how the measurement techniques work (physical measurement principles, hardware requirements, image reconstruction), and 3. how to determine the most suited imaging methods for a given problem.
Literature	Wang, M. (2015). Industrial Tomography. Cambridge, UK: Woodhead Publishing. Available as e-book in the library of TUHH: https://katalog.tub.tuhh.de/Record/823579395

Course L2724: Applied Process Imaging	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Alexander Penn, Dr. Stefan Benders
Language	EN
Cycle	SoSe
Content	Content: The module focuses primarily on discussing established imaging techniques including (a) optical and infrared imaging, (b) magnetic resonance imaging, (c) X-ray imaging and tomography, and (d) ultrasound imaging and also covers a range of more recent imaging modalities. The students will learn: 1. what these imaging techniques can measure (such as sample density or concentration, material transport, chemical composition, temperature), 2. how the measurements work (physical measurement principles, hardware requirements, image reconstruction), and 3. how to determine the most suited imaging methods for a given problem. Learning goals: After the successful completion of the course, the students shall: 1. understand the physical principles and practical aspects of the most common imaging methods, 2. be able to assess the pros and cons of these methods with regard to cost, complexity, expected contrasts, spatial and temporal resolution, and based on this assessment 3. be able to identify the most suited imaging modality for any specific engineering challenge in the field of chemical and bioprocess engineering.
Literature	Wang, M. (2015). Industrial Tomography. Cambridge, UK: Woodhead Publishing. Available as e-book in the library of TUHH: https://katalog.tub.tuhh.de/Record/823579395

Module M2135: Artificial Intelligence				
Courses				
Title	Typ		Hrs/wk	CP
Artificial Intelligence (L3205)	Lecture		2	3
Artificial Intelligence (L3206)	Recitation Section (small)		2	3
Module Responsible	Prof. Pierre-Alexandre Murena			
Admission Requirements	None			
Recommended Previous Knowledge	This course offers an introduction to the domain of Artificial Intelligence beyond Machine Learning. The main objective of the course is therefore to understand other views on AI, to explore problems that data-centric approaches cannot solve directly, and to present some of the technical solutions that have been developed over the decades of AI research. The course will cover in particular logic, knowledge representation and reasoning, knowledge bases... The course concludes with an introduction to neuro-symbolic learning.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i> <i>Skills</i> Personal Competence <i>Social Competence</i> <i>Autonomy</i>				
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written exam			
Examination duration and scale	120 minutes			
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory			

Course L3205: Artificial Intelligence	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Pierre-Alexandre Murena
Language	EN
Cycle	SoSe
Content	This course offers an introduction to the domain of Artificial Intelligence beyond Machine Learning. The main objective of the course is therefore to understand other views on AI, to explore problems that data-centric approaches cannot solve directly, and to present some of the technical solutions that have been developed over the decades of AI research. The course will cover in particular logic, knowledge representation and reasoning, knowledge bases... The course concludes with an introduction to neuro-symbolic learning.
Literature	- Grosan, Crina, et al. "Knowledge representation and reasoning." Intelligent Systems: A Modern Approach (2011): 131-147. - Russell, S. J., & Norvig, P. (2010). Artificial intelligence a modern approach. London. - Bauer, Florian, and Martin Kaltenböck. "Linked open data: The essentials." Edition mono/monochrom, Vienna 710 (2011): 21. - Hitzler, Pascal, and Md Kamruzzaman Sarker, eds. "Neuro-symbolic artificial intelligence: The state of the art." (2022).

Course L3206: Artificial Intelligence	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Pierre-Alexandre Murena
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M2137: TinyML: Embedded Machine Learning				
Courses				
Title	Typ		Hrs/wk	CP
TinyML: Embedded Machine Learning (L3424)	Lecture		2	3
TinyML: Embedded Machine Learning (L3425)	Recitation Section (small)		2	3
Module Responsible	Prof. Olaf Landsiedel			
Admission Requirements	None			
Recommended Previous Knowledge	A student should have taken courses on Internet of Things or embedded systems, such as the course "Internet of Things". Further, we expect students to have taken a course on machine learning and especially neural networks (including knowledge of backpropagation and experience in TensorFlow, PyTorch or similar platforms). Moreover, C programming skills are expected, as we will be using embedded IoT devices.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> - Understanding of the intersection of machine learning and embedded IoT devices. - Familiarity with cutting-edge literature and frameworks in the field of TinyML. - Knowledge of efficient inference techniques, including model compression, pruning, quantization, neural architecture search, and distillation. - Understanding of efficient training techniques, such as distributed training, gradient compression, and on-device transfer learning. - Understanding of how to train and deploy models on microcontrollers. <i>Skills</i> - Practical experience through hands-on project assignments involving embedded AI systems and applications. - Ability to conceive and develop a novel TinyML application running on a microcontroller (MCU). - Application of learned concepts to real-world problems through experimentation and analysis. - C programming skills for use with embedded IoT devices. Personal Competence <i>Social Competence</i> hands-on project assignments promote collaboration and teamwork <i>Autonomy</i> - Ability to independently conceive, design, and develop embedded AI applications. - Capability to understand and tackle fundamental issues in designing methods for embedded AI systems and applications. - Self-driven exploration of cutting-edge literature and frameworks in TinyML to advance understanding and application of the concepts.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	None	Attestation	
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory			

Course L3424: TinyML: Embedded Machine Learning	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Olaf Landsiedel
Language	EN
Cycle	SoSe
Content	Abstract: This course is a course at the intersection of Machine Learning and Embedded IoT Devices. The pervasiveness of ultra-low-power embedded devices, coupled with the introduction of embedded machine learning frameworks, will enable the mass proliferation of AI-powered IoT devices. The rapid growth of machine learning and how easy it is to use machine learning platforms make it an important subject for computer science students to learn about today. Objectives: At the end of the course, students will have learned the following: Practical experience through hands-on project assignments Gained familiarity with cutting-edge literature and frameworks in the field of TinyML Know how to train and deploy models on microcontrollers Conceived and developed a (novel) TinyML application running on a MCU Contents: This course is a deep dive into efficient machine learning techniques that enable powerful deep learning applications on resource-constrained devices. Topics covered • Brief recap / introduction to ML and IoT • efficient inference techniques, including model compression, pruning, quantization, neural architecture search, and distillation; • and efficient training techniques, including distributed training, gradient compression and on-device transfer learning; Our lectures provide students with the required fundamentals, and exercise and projects give students a hands-on experience in developing embedded AI systems & applications and exploring their real-world challenges. This course offers learning experiences that involve hands-on experimentation and analysis as they reinforce student understanding of concepts and their application to real-world problems. Overall, this course provides the students the ability to understand fundamental issues in the design of methods for embedded AI systems and applications. Additional prerequisites: A student should have taken courses on Internet of Things or embedded systems, such as the course "Internet Things and Wireless Networks" at CAU. Further, we expect students to have taken a course on machine learning and especially neural networks (including knowledge of backpropagation and experience in TensorFlow, PyTorch or similar platforms). Moreover, C programming skills are expected, as we will be using embedded IoT devices.
Literature	1.

Course L3425: TinyML: Embedded Machine Learning	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Olaf Landsiedel
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M0623: Intelligent Systems in Medicine				
Courses				
Title		Type	Hrs/wk	CP
Intelligent Systems in Medicine (L0331)		Lecture	2	3
Intelligent Systems in Medicine (L0334)		Project Seminar	2	2
Intelligent Systems in Medicine (L0333)		Recitation Section (small)	1	1
Module Responsible		Prof. Alexander Schlaefer		
Admission Requirements		None		
Recommended Previous Knowledge		• principles of math (algebra, analysis/calculus) • principles of stochastics • principles of programming, Java/C++ and R/Matlab • advanced programming skills		
Educational Objectives		After taking part successfully, students have reached the following learning results		
Professional Competence		<i>Knowledge</i> The students are able to analyze and solve clinical treatment planning and decision support problems using methods for search, optimization, and planning. They are able to explain methods for classification and their respective advantages and disadvantages in clinical contexts. The students can compare different methods for representing medical knowledge. They can evaluate methods in the context of clinical data and explain challenges due to the clinical nature of the data and its acquisition and due to privacy and safety requirements. <i>Skills</i> The students can give reasons for selecting and adapting methods for classification, regression, and prediction. They can assess the methods based on actual patient data and evaluate the implemented methods. Personal Competence <i>Social Competence</i> The students are able to grasp practical tasks in groups, develop solution strategies independently, define work processes and work on them collaboratively. The students can critically reflect on the results of other groups, make constructive suggestions for improvement and also incorporate them into their own work. <i>Autonomy</i> The students can assess their level of knowledge and document their work results. They can critically evaluate the results achieved and present them in an appropriate argumentative manner to the other groups.		
Workload in Hours		Independent Study Time 110, Study Time in Lecture 70		
Credit points		6		
Course achievement		Compulsory	Bonus	Form Description
		Yes	10 %	Written elaboration
		Yes	10 %	Presentation
Examination		Written exam		
Examination duration and scale		90 minutes		
Assignment for the Following Curricula		Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Data Science: Specialisation III. Applications: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Medical Technology: Elective Compulsory Electrical Engineering: Specialisation Medical Technology: Elective Compulsory Interdisciplinary Mathematics: Specialisation Computational Methods in Biomedical Imaging: Compulsory Mechatronics: Core Qualification: Elective Compulsory Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory Biomedical Engineering: Specialisation Implants and Endoprotheses: Elective Compulsory Biomedical Engineering: Specialisation Management and Business Administration: Elective Compulsory Biomedical Engineering: Specialisation Medical Technology and Control Theory: Compulsory Theoretical Mechanical Engineering: Specialisation Bio- and Medical Technology: Elective Compulsory		

Course L0331: Intelligent Systems in Medicine	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Alexander Schläefer
Language	EN
Cycle	WiSe
Content	- methods for search, optimization, planning, classification, regression and prediction in a clinical context - representation of medical knowledge - understanding challenges due to clinical and patient related data and data acquisition The students will work in groups to apply the methods introduced during the lecture using problem based learning.
Literature	Russel & Norvig: Artificial Intelligence: a Modern Approach, 2012 Berner: Clinical Decision Support Systems: Theory and Practice, 2007 Greenes: Clinical Decision Support: The Road Ahead, 2007 Further literature will be given in the lecture

Course L0334: Intelligent Systems in Medicine	
Typ	Project Seminar
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Alexander Schläefer
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Course L0333: Intelligent Systems in Medicine	
Typ	Recitation Section (small)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Alexander Schläefer
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1249: Medical Imaging					
Courses					
Title		Typ	Hrs/wk	CP	
Medical Imaging (L1694)		Lecture	2	3	
Medical Imaging (L1695)		Recitation Section (small)	2	3	
Module Responsible	Prof. Tobias Knopp				
Admission Requirements	None				
Recommended Previous Knowledge	Basic knowledge in linear algebra, numerics, and signal processing				
Educational Objectives	After taking part successfully, students have reached the following learning results				
Professional Competence	Knowledge	After successful completion of the module, students are able to describe reconstruction methods for different tomographic imaging modalities such as computed tomography and magnetic resonance imaging. They know the necessary basics from the fields of signal processing and inverse problems and are familiar with both analytical and iterative image reconstruction methods. The students have a deepened knowledge of the imaging operators of computed tomography and magnetic resonance imaging.			
		Skills	The students are able to implement reconstruction methods and test them using tomographic measurement data. They can visualize the reconstructed images and evaluate the quality of their data and results. In addition, students can estimate the temporal complexity of imaging algorithms.		
	Personal Competence		Social Competence	Students can work on complex problems both independently and in teams. They can exchange ideas with each other and use their individual strengths to solve the problem.	
				Autonomy	Students are able to independently investigate a complex problem and assess which competencies are required to solve it.
	Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6				
Course achievement	None				
Examination	Written exam				
Examination duration and scale	90 min				
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Data Science: Specialisation III. Applications: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Medical Technology: Elective Compulsory Electrical Engineering: Specialisation Medical Technology: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Interdisciplinary Mathematics: Specialisation Computational Methods in Biomedical Imaging: Compulsory Microelectronics and Microsystems: Specialisation Communication and Signal Processing: Elective Compulsory Technomathematics: Specialisation II. Informatics: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Bio- and Medical Technology: Elective Compulsory				

Course L1694: Medical Imaging	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Tobias Knopp
Language	EN
Cycle	WiSe
Content	• Overview about different imaging methods • Signal processing • Inverse problems • Computed tomography • Magnetic resonance imaging • Compressed Sensing • Magnetic particle imaging
Literature	Bildgebende Verfahren in der Medizin; O. Dössel; Springer, Berlin, 2000 Bildgebende Systeme für die medizinische Diagnostik; H. Morneburg (Hrsg.); Publicis MCD, München, 1995 Introduction to the Mathematics of Medical Imaging; C. L.Epstein; Siam, Philadelphia, 2008 Medical Image Processing, Reconstruction and Restoration; J. Jan; Taylor and Francis, Boca Raton, 2006 Principles of Magnetic Resonance Imaging; Z.-P. Liang and P. C. Lauterbur; IEEE Press, New York, 1999

Course L1695: Medical Imaging	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Tobias Knopp
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1961: Architectures and Algorithms for Deep-Learning Acceleration				
Courses				
Title		Typ	Hrs/wk	CP
Architectures and Algorithms for Deep-Learning Acceleration (L3202)		Lecture	2	3
Architectures and Algorithms for Deep-Learning Acceleration (L3203)		Recitation Section (small)	2	3
Module Responsible	Prof. Görschwin Fey			
Admission Requirements	None			
Recommended Previous Knowledge	• Computer Architecture • Computer Engineering • Basic Programming Skills • Linear Algebra			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> The students acquire specialized knowledge about the architectures & algorithms that empower IoT and other embedded devices to perform computationally complex deep-learning tasks, originally only suitable for powerful server devices. <i>Skills</i> Thereby, the students will not only learn the structure of modern deep-learning networks for computer vision and natural language processing, but also how to build and train their own networks based on the constraints of the target system (e.g., a low memory footprint). Moreover, the students will learn how modern domain-specific architectures for deep-learning acceleration are architected and implemented. Personal Competence <i>Social Competence</i> Students are able to solve similar problems alone or in a group and to present the results accordingly. <i>Autonomy</i> Students are able to acquire new knowledge from specific literature and to associate this knowledge with other classes.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Oral exam			
Examination duration and scale	20 min			
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory			

Course L3202: Architectures and Algorithms for Deep-Learning Acceleration	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Lennart Bamberg
Language	EN
Cycle	WiSe
Content	• Fundamentals of deep learning • Applications of deep learning in edge/embedded devices • Implementation and training of neural networks using a SOTA Python framework (e.g., Tensorflow) • Deployment of SOTA neural networks on embedded devices • Methodologies for improved deep-learning inference on constrained devices (e.g., IoT nodes and embedded systems) • Hardware architectures for improved deep-learning inference on constrained devices (e.g., IoT nodes and embedded systems) • Introduction to neuromorphic/bio-inspired computer architectures
Literature	Zhang, Aston, et al. "Dive into deep learning." (2021). Online available for free under https://d2l.ai and https://arxiv.org/ftp/arxiv/papers/2106/2106.11342.pdf

Course L3203: Architectures and Algorithms for Deep-Learning Acceleration	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Lennart Bamberg
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1748: Construction Robotics				
Courses				
Title	Typ		Hrs/wk	CP
Construction Robotics (L2867)	Project-/problem-based Learning		6	6
Module Responsible	Prof. Kay Smarsly			
Admission Requirements	None			
Recommended Previous Knowledge	Basics of project-oriented programming			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	Basics of robotics Applications in civil engineering Kinematics Use of specific hardware Development of software routines Python programming language Image processing Basics of localization (LIDAR, SLAM)			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>				
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84			
Credit points	6			
Course achievement	None			
Examination	Written elaboration			
Examination duration and scale	ca. 10 Seiten			
Assignment for the Following Curricula	Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Civil Engineering: Specialisation Coastal Engineering: Elective Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory Civil Engineering: Specialisation Computational Engineering: Elective Compulsory Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L2867: Construction Robotics	
Typ	Project-/problem-based Learning
Hrs/wk	6
CP	6
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84
Lecturer	Prof. Kay Smarsly, Jan Stührenberg
Language	EN
Cycle	WiSe
Content	1. Introduction: Robotics in civil engineering 2. Presentation of potential topics 3. Programming of algorithms in Python 4. Application of software systems: LINUX distribution, ROS, CloudCompare, ... 5. Application of hardware systems: Petoï Bittle Dog, Raspberry Pi, Arduino, sensing ... 6. Topics considered for robotics using the Petoï Bittle Dog: 1. Movement 2. Use of sensors (camera, infrared, ...) 3. Data structures/data acquisition 4. Programming 7. Topics technically relevant to building inspection: 1. Geodetic evaluations 2. Image processing 3. Localization
Literature	Bock/Linner: Construction Robotics Verl et al.: Soft Robotics Pasquale: New Laws of robotics

Module M2001: Reinforcement Learning				
Courses				
Title	Typ		Hrs/wk	CP
Reinforcement Learning (L3305)	Lecture		2	3
Reinforcement Learning (L3306)	Project-/problem-based Learning		2	3
Module Responsible	Dr. Manfred Eppe			
Admission Requirements	None			
Recommended Previous Knowledge	- Lectures at TUHH: - Machine Learning I and II - Advanced Machine Learning - Programming Paradigms - Additional recommended knowledge and prerequisites: - Python programming, including libraries for neural networks, deep learning and data visualization (e.g. wandb). - Access to a decent PC or laptop with at least mediocre performance.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i>	- Foundations of Reinforcement Learning, including - Markov Decision Processes - Dynamic Programming - Value- and Policy Iteration - TD-Learning and MC-Methods - Policy Gradient Methods - Q-Learning - Actor-Critic Methods - Individual advanced competences, at least one of the following: - Soft Actor-Critic - Proximal Policy Optimization - Multi-Agent RL - Goal-Conditioned RL - Intrinsic Rewards and Reward Shaping - Hierarchical RL - Model-based RL			
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>	- Working in a team - Conflict resolution in a team - Coordination of tasks in a team			
	- Improvement of English writing and speaking - Scientific work - Presentation skills			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	20 %	Presentation	Projektpräsentation
Examination	Oral exam			
Examination duration and scale	30 min			
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory			

Course L3305: Reinforcement Learning	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Manfred Eppe, Frank Röder
Language	EN
Cycle	WiSe
Content	How can robotic and artificial agents learn to act appropriately in their environment? This course conveys foundational and advanced concepts of Reinforcement Learning to address this question. The core concepts will be taught in a lecture. In addition, there will be a strong focus on practical project work in the seminar part of this course. The practical applications will involve virtual agents in simulated 2d- or 3d environments. Prerequisites for participation are knowledge in machine learning, neural networks, stochastics, Python programming and a decent own PC or laptop with at least mediocre performance. The corresponding project/problem-based course complements the theoretical content of the lecture with a specific technical problem (mini-projects) that students solve in group work throughout the semester. The topics come from the field of robotics and are solved using 3D simulations in our institute's virtual robotics laboratory (Scilab-RL).
Literature	Reinforcement Learning: An Introduction (Sutton & Barto) Wiki and website of our virtual robotics lab (Scilab-RL): https://scilab-rl.github.io/Scilab-RL/

Course L3306: Reinforcement Learning	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Manfred Eppe, Frank Röder
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Specialization III. Mathematics

Module M0667: Algorithmic Algebra

Courses

Title	Typ	Hrs/wk	CP
Algorithmic Algebra (L0422)	Lecture	2	4
Algorithmic Algebra (L0423)	Recitation Section (small)	2	2
Module Responsible	Dr. Prashant Batra		
Admission Requirements	None		
Recommended Previous Knowledge	Mathe I-III (Real analysis, computing in Vector spaces, principle of complete induction) Diskrete Mathematik I (groups, rings, ideals, fields; euclidean algorithm)		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence			
<i>Knowledge</i>	Students can name the basic terms, structures and questions of algorithmic algebra and explain them using examples, such as: Fast arithmetic, prime number tests and factorizability, Diophantine equations in commutative rings, calculation of Gröbner bases, algorithmic resolution of multivariate systems of equations. Students are familiar with paradigms for designing fast algorithms and complexity theory analysis.		
<i>Skills</i>	Students are able to design fast algebraic algorithms. Students can develop a suitable solution approach for given problems, pursue it and critically evaluate the results, for example when solving multivariate systems of equations and in lattice point theory. Students can model tasks from algorithmic algebra using the concepts presented and solve them using the methods learned: in particular (non)-linear Diophantine equations over rings, systems of equations over rings, Bezout equations in matrix rings. Students are able to independently discover further logical connections between the concepts they have learned and can verify them.		
Personal Competence			
<i>Social Competence</i>	Students are able to work together in teams and master mathematics as a common language.		
<i>Autonomy</i>	Students are able to discuss their own algorithmic suggestions for solving the problems presented with other students. Students learn to present their suggestions coherently to a group.		
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Oral exam		
Examination duration and scale	30 min		
Assignment for the Following Curricula	Computer Science: Specialisation III. Mathematics: Elective Compulsory		

Course L0422: Algorithmic Algebra

Typ	Lecture
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Dr. Prashant Batra
Language	DE/EN
Cycle	WiSe
Content	fast arithmetic algorithms (conversion, fast multiplications) discrete Fourier-transformation over rings Division with remainder (over rings) Extended euclidean algorithm, solution of the Bezout-equation Computing with modular remainders, solving of remainder systems (chinese remainder theorem), solvability of integer linear systems over the integers Tests for primeness, factorization, fast computing with congruence classes for RSA Linearization of polynomial equations -- matrix approach Sylvester-matrix (to detect non-trivial gcd's), elimination Elimination in rings, elimination of many variables Buchberger algorithm, Gröbner basis

	Minkowskis Lattice Point theorem and integer-valued optimization LLL-algorithm for construction of 'short' lattice vectors in polynomial time
Literature	von zur Gathen, Joachim; Gerhard, Jürgen: Modern computer algebra. 3rd ed. Cambridge: Cambridge University Press, 2013. http://dx.doi.org/10.1017/CBO9781139856065 Yap, Chee Keng: Fundamental problems of algorithmic algebra. Oxford University Press. Oxford, 2000. Free download for students from author's website: https://cs.nyu.edu/cs/faculty/yap/book/berlin/ Lauritzen: Niels: Concrete abstract algebra : from numbers to Gröbner bases. Cambridge Univ. Press, 2006. ISBN 978-0-521-53410-9. Signatur MAD-341 in der TUHH-Lehrbuchsammlung Cox, David; Little, John; O'Shea, Donal Ideals, varieties, and algorithms. An introduction to computational algebraic geometry and commutative algebra. 3rd ed. New York, NY: Springer 551 p. eBook: http://dx.doi.org/10.1007/978-0-387-35651-8 Koepf, Wolfram: Computeralgebra. Eine algorithmisch orientierte Einführung. Berlin: Springer (ISBN 3-540-29894-0/pbk). xiii, 515 p. springer eBook: http://dx.doi.org/10.1007/3-540-29895-9 Kaplan, Michael: Computer algebra. (German) Berlin: Springer (ISBN 3-540-21379-1/pbk). xii, 391 p. springer eBook: http://dx.doi.org/10.1007/b137968 L.N. Childs: A concrete introduction to higher algebra; 3ed, Springer, 2009. https://link.springer.com/book/10.1007/978-0-387-74725-5 Karpfinger, Christian; Kiechle, Hubert : Kryptologie- Algebraische Methoden und Algorithmen. Vieweg+Teubner, 2010. e-book: https://doi.org/10.1007/978-3-8348-9356-7 Alin Bostan, Frédéric Chyzak, Marc Giusti, Romain Lebreton, Grégoire Lecerf, Bruno Salvy et Éric Schost. <i>Algorithmes Efficaces en Calcul Formel</i>. 686 pages. Imprimé par CreateSpace. Palaiseau : Frédéric Chyzak (auto-édit.), sept. 2017. isbn : 979-10-699-0947-2. url : https://hal.archives-ouvertes.fr/AECF/. Free e-book: https://mathexp.eu/chyzak/aecf-distrib/pdf/aecf-screen-1.1.pdf,

Course L0423: Algorithmic Algebra	
Typ	Recitation Section (small)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Prashant Batra
Language	DE/EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1428: Linear and Nonlinear Optimization				
Courses				
Title	Typ		Hrs/wk	CP
Linear and Nonlinear Optimization (L2062)	Lecture		4	4
Linear and Nonlinear Optimization (L2063)	Recitation Section (large)		1	2
Module Responsible	Prof. Matthias Mnich			
Admission Requirements	None			
Recommended Previous Knowledge	- Discrete Algebraic Structures - Mathematics I - Graph Theory and Optimization			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i> - Students can name the basic concepts in linear and non-linear optimization. They are able to explain them using appropriate examples. - Students can discuss logical connections between these concepts. They are capable of illustrating these connections with the help of examples. - They know proof strategies and can reproduce them. <i>Skills</i> - Students can model problems in linear and non-linear optimization with the help of the concepts studied in this course. Moreover, they are capable of solving them by applying established methods. - Students are able to discover and verify further logical connections between the concepts studied in the course. - For a given problem, the students can develop and execute a suitable approach, and are able to critically evaluate the results. Personal Competence <i>Social Competence</i> - Students are able to work together in teams. They are capable to use mathematics as a common language. - In doing so, they can communicate new concepts according to the needs of their cooperating partners. Moreover, they can design examples to check and deepen the understanding of their peers. <i>Autonomy</i> - Students are capable of checking their understanding of complex concepts on their own. They can specify open questions precisely and know where to get help in solving them. - Students have developed sufficient persistence to be able to work for longer periods in a goal-oriented manner on hard problems.				
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	20 %	Exercises	
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Computer Science in Engineering: Specialisation III. Mathematics: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Energy Systems: Elective Compulsory			

Course L2062: Linear and Nonlinear Optimization	
Typ	Lecture
Hrs/wk	4
CP	4
Workload in Hours	Independent Study Time 64, Study Time in Lecture 56
Lecturer	Prof. Matthias Mnich
Language	DE/EN
Cycle	WiSe
Content	• Modelling linear programming problems • Graphical method • Algebraic background • Convexity • Polyhedral theory • Simplex method • Degeneracy and convergence • duality • interior-point methods • quadratic optimization • integer linear programming
Literature	• A. Schrijver: Combinatorial Optimization: Polyhedra and Efficiency. Springer, 2003 • B. Korte and T. Vygen: Combinatorial Optimization: Theory and Algorithms. Springer, 2018 • T. Cormen, Ch. Leiserson, R. Rivest, C. Stein: Introduction to Algorithms. MIT Press, 2013

Course L2063: Linear and Nonlinear Optimization	
Typ	Recitation Section (large)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Matthias Mnich
Language	DE/EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0716: Hierarchical Algorithms				
Courses				
Title	Typ		Hrs/wk	CP
Hierarchical Algorithms (L0585)	Lecture		2	3
Hierarchical Algorithms (L0586)	Recitation Section (small)		2	3
Module Responsible	Prof. Sabine Le Borne			
Admission Requirements	None			
Recommended Previous Knowledge	- Mathematics I, II, III for Engineering students (german or english) or Analysis & Linear Algebra I + II as well as Analysis III for Technomathematicians - Numerical Mathematics 1 - Programming experience in C			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> Students are able to - name representatives of hierarchical algorithms and list their characteristics, - explain construction techniques for hierarchical algorithms, - discuss aspects regarding the efficient implementation of hierarchical algorithms. <i>Skills</i> Students are able to - implement the hierarchical algorithms discussed in the lecture, - analyse the storage and computational complexities of the algorithms, - adapt algorithms to problem settings of various applications and thus develop problem adapted variants. Personal Competence <i>Social Competence</i> Students are able to - work together in heterogeneously composed teams (i.e., teams from different study programs and background knowledge), explain theoretical foundations and support each other with practical aspects regarding the implementation of algorithms. <i>Autonomy</i> Students are capable - to assess whether the supporting theoretical and practical exercises are better solved individually or in a team, - to work on complex problems over an extended period of time, - to assess their individual progress and, if necessary, to ask questions and seek help.			
Workload in Hours				
Credit points				
Course achievement				
Examination				
Examination duration and scale				
Assignment for the Following Curricula	Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Technomathematics: Specialisation I. Mathematics: Elective Compulsory Technomathematics: Specialisation II. Informatics: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Simulation Technology: Elective Compulsory			

Course L0585: Hierarchical Algorithms	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Sabine Le Borne
Language	EN
Cycle	WiSe
Content	- Low rank matrices - Separable expansions - Hierarchical matrix partitions - Hierarchical matrices - Formatted matrix operations - H2 matrices - Applications - Additional topics (e.g. matrix functions, tensor products)
Literature	W. Hackbusch: Hierarchical matrices: Algorithms and Analysis, Springer (2015)

Course L0586: Hierarchical Algorithms	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Sabine Le Borne
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M2071: Linear Models and Regression			
Courses			
Title	Typ	Hrs/wk	CP
Linear Models and Regression (L3343)	Lecture	3	4
Linear Models and Regression (L3344)	Recitation Section (small)	1	2
Module Responsible	Prof. Matthias Schulte		
Admission Requirements	None		
Recommended Previous Knowledge	Preknowledge in probability and statistics		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i> - Students know the fundamental models for regression and corresponding tests and are able to explain them using appropriate examples. - Students can discuss logical connections between these concepts and are capable of illustrating these connections with the help of examples. - Students know proof strategies and can reproduce them. <i>Skills</i> - Students can investigate statistical problems with the help of the models studied in the course. - Students are able to explore and verify further logical connections between the concepts studied in the course. - For a given problem, the students can develop and execute a suitable approach, and are able to critically evaluate the results. Personal Competence <i>Social Competence</i> - Students are able to work together (e.g. on their regular home work) and to present their results appropriately (e.g. during exercise class). - In doing so, they can communicate new concepts and they can design examples to check and deepen the understanding of their peers. <i>Autonomy</i> - Students are capable of checking their understanding of complex concepts on their own. They can specify open questions precisely and know where to get help in solving them. - Students can put their knowledge in relation to the contents of other lectures. - Students have developed sufficient persistence to be able to work for longer periods in a goal-oriented manner on hard problems.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	90 min		
Assignment for the Following Curricula	Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Core Qualification: Compulsory Computer Science in Engineering: Specialisation III. Mathematics: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory		

Course L3343: Linear Models and Regression	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Matthias Schulte
Language	EN
Cycle	WiSe
Content	- Linear models - Testing in the Gaussian linear model - Analysis of variance - Asymptotic behaviour of maximum likelihood estimators - Nonlinear regression - Logistic and Poisson regression - Generalised linear models - Linear discriminant analysis - Principal component analysis
Literature	P. Dunn and G. Smyth (2018): Generalized linear models with examples in R. Springer. L. Fahrmeir, T. Kneib, S. Lang and B. Marx (2021): Regression - models, methods and applications. Second edition, Springer.

Course L3344: Linear Models and Regression	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Matthias Schulte
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M2090: Quantum Mechanics				
Courses				
Title	Typ		Hrs/wk	CP
Quantum Mechanics (L3363)	Lecture		2	3
Quantum Mechanics (L3364)	Recitation Section (large)		1	1
Quantum Mechanics (L3382)	Recitation Section (small)		2	2
Module Responsible	Prof. Martin Kliesch			
Admission Requirements	None			
Recommended Previous Knowledge	- Knowledge of mathematics is required, including linear algebra, vector calculus, complex numbers, the Fourier transform, and wave equations. - Some Knowledge in physics is helpful: wave phenomena, harmonic oscillators (classical), and electromagnetism.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i>	This module is an introduction to quantum mechanics on the master level. The students learn basic paradigmatic models of quantum mechanics and acquire the ability to describe and explain the basic terms and principles. They can distinguish similarities and differences between quantum mechanics and classical physics and know in which situations quantum mechanical phenomena may be expected. An emphasis will be placed on conceptual and mathematical aspects.			
<i>Skills</i> Personal Competence <i>Social Competence</i>	After completing this module, students can - reproduce the knowledge taught in the course, - reproduce simpler proofs of the course and reproduce the ideas of the more complicated ones, - establish connections between the concepts taught and - apply the learned knowledge to concrete problems. After completing this module, students are expected to be able to work on subject-specific tasks alone or in a group and to present the results appropriately.			
<i>Autonomy</i> Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	15 %	Exercises	
Examination	Written exam			
Examination duration and scale	120 min			
Assignment for the Following Curricula	Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Microelectronics and Microsystems: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L3363: Quantum Mechanics	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Martin Kliesch
Language	EN
Cycle	WiSe
Content	• Overview of experiments (Stern-Gerlach, black body radiation, photoelectric effect, double slit experiment) • Wave equations & dispersion relations • Schrödinger equation, wave packets, tunnelling effect • Postulates of quantum mechanics (pure states) • Harmonic oscillator • Heisenberg uncertainty relation • Hydrogen Atom • Composite quantum systems • Spin & coupling to magnetic fields • Outlook: Fermions & Bosons
Literature	• Sakurai and Napolitano, Modern Quantum Mechanics (3rd ed., 2020) • Nielsen and Chuang, Quantum Computation and Quantum Information

Course L3364: Quantum Mechanics	
Typ	Recitation Section (large)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Martin Kliesch
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Course L3382: Quantum Mechanics	
Typ	Recitation Section (small)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Martin Kliesch
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1405: Randomised Algorithms and Random Graphs			
Courses			
Title	Typ	Hrs/wk	CP
Randomised Algorithms and Random Graphs (L2010)	Lecture	2	3
Randomised Algorithms and Random Graphs (L2011)	Recitation Section (large)	2	3
Module Responsible	Prof. Anusch Taraz		
Admission Requirements	None		
Recommended Previous Knowledge			
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i> - Students can describe basic concepts in the area of Randomized Algorithms and Random Graphs such as random walks, tail bounds, fingerprinting and algebraic techniques, first and second moment methods, and various random graph models. They are able to explain them using appropriate examples. - Students can discuss logical connections between these concepts. They are capable of illustrating these connections with the help of examples. - They know proof strategies and can apply them. <i>Skills</i> - Students can model problems with the help of the concepts studied in this course. Moreover, they are capable of solving them by applying established methods. - Students are able to explore and verify further logical connections between the concepts studied in the course. - For a given problem, the students can develop and execute a suitable technique, and are able to critically evaluate the results. Personal Competence <i>Social Competence</i> - Students are able to work together in teams. They are capable to establish a common language. - In doing so, they can communicate new concepts according to the needs of their cooperating partners. Moreover, they can design examples to check and deepen the understanding of their peers. <i>Autonomy</i> - Students are capable of checking their understanding of complex concepts on their own. They can specify open questions precisely and know where to get help in solving them. - Students have developed sufficient persistence to be able to work for longer periods in a goal-oriented manner on hard problems.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Oral exam		
Examination duration and scale	30 min		
Assignment for the Following Curricula	Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Computer Science in Engineering: Specialisation III. Mathematics: Elective Compulsory		

Course L2010: Randomised Algorithms and Random Graphs	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Anusch Taraz, Prof. Volker Turau
Language	DE/EN
Cycle	SoSe
Content	Randomized Algorithms: • introduction and recalling basic tools from probability • randomized search • random walks • text search with fingerprinting • parallel and distributed algorithms • online algorithms Random Graphs: • typical properties • first and second moment method • tail bounds • thresholds and phase transitions • probabilistic method • models for complex networks
Literature	• Motwani, Raghavan: Randomized Algorithms • Worsch: Randomisierte Algorithmen • Dietzfelbinger: Randomisierte Algorithmen • Bollobas: Random Graphs • Alon, Spencer: The Probabilistic Method • Frieze, Karonski: Random Graphs • van der Hofstad: Random Graphs and Complex Networks

Course L2011: Randomised Algorithms and Random Graphs	
Typ	Recitation Section (large)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Anusch Taraz, Prof. Volker Turau
Language	DE/EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M0714: Numerical Methods for Ordinary Differential Equations			
Courses			
Title	Typ	Hrs/wk	CP
Numerical Treatment of Ordinary Differential Equations (L0576)	Lecture	2	3
Numerical Treatment of Ordinary Differential Equations (L0582)	Recitation Section (small)	2	3
Module Responsible	Prof. Daniel Ruprecht		
Admission Requirements	None		
Recommended Previous Knowledge	- Mathematik I, II, III for Engineers (German or English) or Analysis & Linear Algebra I + II plus Analysis III for Technomathematiker. - Basic knowledge of MATLAB, Python or a similar programming language.		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	<i>Knowledge</i> Students are able to - name numerical methods for the solution of ordinary differential equations and explain their core ideas, - formulate convergence statements for the taught numerical methods (including the necessary assumptions about the solved problem), - explain aspects regarding the practical realisation of a method, - select the appropriate numerical method for specific problems, implement the numerical algorithms efficiently and interpret the numerical results. <i>Skills</i> Students are able to - implement, apply and compare numerical methods for the solution of ordinary differential equations, - explain the convergence behaviour of numerical methods, taking into consideration the solved problem and selected algorithm, - develop a suitable solution approach for a given problem, if necessary by combining multiple algorithms, realise this approach and critically evaluate results. Personal Competence <i>Social Competence</i> Students are able to - work together in heterogeneous teams (i.e., teams from different study programs and with different background knowledge), explain theoretical foundations and support each other with practical aspects regarding the implementation of algorithms. <i>Autonomy</i> Students are capable - to assess whether the provided theoretical and practical exercises are better solved individually or in a team and - to assess their individual progress and, if necessary, to ask questions and seek help.		
Workload in Hours			
Credit points			
Course achievement			
Examination			
Examination duration and scale			
Assignment for the Following Curricula	Bioprocess Engineering: Specialisation A - General Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Technical Complementary Course: Elective Compulsory Chemical and Bioprocess Engineering: Technical Complementary Course: Elective Compulsory Computational Engineering: Core Qualification: Elective Compulsory Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Elective Compulsory Electrical Engineering: Specialisation Control and Power Systems Engineering: Elective Compulsory Energy Systems: Core Qualification: Elective Compulsory Aircraft Systems Engineering: Core Qualification: Elective Compulsory Interdisciplinary Mathematics: Specialisation II. Numerical - Modelling Training: Compulsory Mechatronics: Core Qualification: Elective Compulsory Technomathematics: Specialisation I. Mathematics: Elective Compulsory Theoretical Mechanical Engineering: Core Qualification: Compulsory Process Engineering: Specialisation Chemical Process Engineering: Elective Compulsory Process Engineering: Specialisation Process Engineering: Elective Compulsory		

Course L0576: Numerical Treatment of Ordinary Differential Equations	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Daniel Ruprecht
Language	DE/EN
Cycle	SoSe
Content	Numerical methods for Initial Value Problems • single step methods • multistep methods • stiff problems • differential algebraic equations (DAE) of index 1 Numerical methods for Boundary Value Problems • multiple shooting method • difference methods
Literature	• E. Hairer, S. Nørsett, G. Wanner: Solving Ordinary Differential Equations I: Nonstiff Problems. • E. Hairer, G. Wanner: Solving Ordinary Differential Equations II: Stiff and Differential-Algebraic Problems. • D. Griffiths, D. Higham: Numerical Methods for Ordinary Differential Equations.

Course L0582: Numerical Treatment of Ordinary Differential Equations	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Daniel Ruprecht
Language	DE/EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1668: Probability Theory				
Courses				
Title	Typ		Hrs/wk	CP
Probability Theory (L2643)	Lecture		3	4
Probability Theory (L2644)	Recitation Section (small)		1	2
Module Responsible	Prof. Matthias Schulte			
Admission Requirements	None			
Recommended Previous Knowledge	Familiarity with the basic concepts of probability			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i> - Students can name the basic concepts in probability theory. They are able to explain them using appropriate examples. - Students can discuss logical connections between these concepts. They are capable of illustrating these connections with the help of examples. - They know proof strategies and can reproduce them. <i>Skills</i> - Students can model problems from probability theory with the help of the concepts studied in this course. Moreover, they are capable of solving them by applying established methods. - Students are able to explore and verify further logical connections between the concepts studied in the course. - For a given problem, the students can develop and execute a suitable technique, and are able to critically evaluate the results. Personal Competence <i>Social Competence</i> - Students are able to work together (e.g. on their regular home work) and to present their results appropriately (e.g. during exercise class). - In doing so, they can communicate new concepts according to the needs of their cooperating partners. Moreover, they can design examples to check and deepen the understanding of their peers. <i>Autonomy</i> - Students are capable of checking their understanding of complex concepts on their own. They can specify open questions precisely and know where to get help in solving them. - Students can put their knowledge in relation to the contents of other lectures. - Students have developed sufficient persistence to be able to work for longer periods in a goal-oriented manner on hard problems.				
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	5 %	Exercises	
Examination	Oral exam			
Examination duration and scale	30 min			
Assignment for the Following Curricula	Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Interdisciplinary Mathematics: Specialisation II. Numerical - Modelling Training: Elective Compulsory Technomathematics: Specialisation I. Mathematics: Elective Compulsory			

Course L2643: Probability Theory	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Matthias Schulte
Language	EN
Cycle	SoSe
Content	• Measure and probability spaces • Integration and expectation • Types of stochastic convergence • Law of large numbers • Central limit theorem • Radon-Nikodym theorem • Conditional expectation • Martingales • Markov chains • Poisson processes
Literature	H. Bauer, Probability theory and elements of measure theory, second edition, Academic Press, 1981. A. Klenke, Probability Theory: A Comprehensive Course, second edition, Springer, 2014. G. F. Lawler, Introduction to Stochastic Processes, second edition, Chapman & Hall/CRC, 2006. A. N. Shiryaev, Probability, second edition, Springer, 1996.

Course L2644: Probability Theory	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Matthias Schulte
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M0711: Numerical Mathematics II			
Courses			
Title	Typ	Hrs/wk	CP
Numerical Mathematics II (L0568)	Lecture	2	3
Numerical Mathematics II (L0569)	Recitation Section (small)	2	3
Module Responsible	Dr. Jens-Peter Zemke		
Admission Requirements	None		
Recommended Previous Knowledge	- Numerical Mathematics I - Python knowledge		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i>	Students are able to - name advanced numerical methods for interpolation, approximation, integration, eigenvalue problems, eigenvalue problems, nonlinear root finding problems and explain their core ideas, - repeat convergence statements for the numerical methods, sketch convergence proofs, - explain practical aspects of numerical methods concerning runtime and storage needs - explain aspects regarding the practical implementation of numerical methods with respect to computational and storage complexity.		
<i>Skills</i>	Students are able to - implement, apply and compare advanced numerical methods in Python, - justify the convergence behaviour of numerical methods with respect to the problem and solution algorithm and to transfer it to related problems, - for a given problem, develop a suitable solution approach, if necessary through composition of several algorithms, to execute this approach and to critically evaluate the results		
Personal Competence <i>Social Competence</i>	Students are able to work together in heterogeneously composed teams (i.e., teams from different study programs and background knowledge), explain theoretical foundations and support each other with practical aspects regarding the implementation of algorithms.		
<i>Autonomy</i>	Students are capable - to assess whether the supporting theoretical and practical exercises are better solved individually or in a team, - to assess their individual progress and, if necessary, to ask questions and seek help.		
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Oral exam		
Examination duration and scale	25 min		
Assignment for the Following Curricula	Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Energy Systems: Core Qualification: Elective Compulsory Computer Science in Engineering: Specialisation III. Mathematics: Elective Compulsory Technomathematics: Specialisation I. Mathematics: Elective Compulsory Theoretical Mechanical Engineering: Core Qualification: Elective Compulsory		

Course L0568: Numerical Mathematics II	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Jens-Peter Zemke
Language	DE/EN
Cycle	SoSe
Content	1. Error and stability: Notions and estimates 2. Rational interpolation and approximation 3. Multidimensional interpolation (RBF) and approximation (neural nets) 4. Quadrature: Gaussian quadrature, orthogonal polynomials 5. Linear systems: Perturbation theory of decompositions, structured matrices 6. Eigenvalue problems: LR-, QD-, QR-Algorithmus 7. Nonlinear systems of equations: Newton and Quasi-Newton methods, line search (optional) 8. Krylov space methods: Arnoldi-, Lanczos methods (optional)
Literature	• Skript • Stoer/Bulirsch: Numerische Mathematik 1, Springer • Dahmen, Reusken: Numerik für Ingenieure und Naturwissenschaftler, Springer

Course L0569: Numerical Mathematics II	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Jens-Peter Zemke
Language	DE/EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1965: Mathematics of Constraint Satisfaction				
Courses				
Title		Typ	Hrs/wk	CP
Mathematics of Constraint Satisfaction (L3209)		Lecture	2	3
Mathematics of Constraint Satisfaction (L3210)		Recitation Section (small)	2	3
Module Responsible	Prof. Antoine Wiehe			
Admission Requirements	None			
Recommended Previous Knowledge	The students should have followed the courses Computability and Complexity Theory, Discrete Algebraic Structures, as well as Logic in Computer Science.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i> <i>Skills</i>				
Personal Competence <i>Social Competence</i> <i>Autonomy</i>				
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Oral exam			
Examination duration and scale	30 min			
Assignment for the Following Curricula	Computer Science: Specialisation III. Mathematics: Elective Compulsory Technomathematics: Specialisation II. Informatics: Elective Compulsory			

Course L3209: Mathematics of Constraint Satisfaction	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Antoine Wiehe
Language	EN
Cycle	SoSe
Content	This course gives an introduction to the topic of constraint satisfaction problems and their complexity. A constraint satisfaction problem (CSP) is a computational problem of the form "Given variables and constraints on the variables, does there exist an assignment of the variables to some concrete domain that satisfies all the constraints?" The framework of CSPs is very general and captures a variety of problems from boolean satisfiability to graph coloring problems, including linear programming or even solving Sudokus. In fact, every computational problem is equivalent to a CSP. The study of CSPs has been very prolific in the past, both in practice (e.g., with SAT solvers) and in complexity theory, a prominent field of theoretical computer science. The research of the last three decades has shown an intimate connection between CSPs and various fields of mathematics, in particular with universal algebra. In this course, we will review the theoretical aspects of CSPs. The course will cover the basics of the theory such as the universal-algebraic approach to constraint satisfaction and several classical algorithms such as local consistency checking and the Bulatov-Dalmau algorithm. Basic knowledge in predicate logic and an affinity to abstract mathematical thinking are highly recommended in order to follow this course.
Literature	• A script for the lecture • Polymorphisms and How to Use Them, Barto, Krokhin, Willard.

Course L3210: Mathematics of Constraint Satisfaction	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Antoine Wiehe
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M2086: Advanced Quantum Computing				
Courses				
Title	Typ		Hrs/wk	CP
Advanced Quantum Computing (L3249)	Lecture		2	3
Advanced Quantum Computing (L3250)	Recitation Section (small)		2	3
Module Responsible	Prof. Martin Kliesch			
Admission Requirements	None			
Recommended Previous Knowledge	1. Linear algebra and very good mathematical skills 2. Skills from the course "Introduction to quantum computing" The skills for 2. can alternatively be obtained in the seminar "Quantum mechanics as a generalization of classical probability theory." As a rough guideline, the participants must understand all the details of the quantum teleportation protocol as a prerequisite.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> Quantum computing is expected to solve relevant computational problems faster than classical computers. Currently, major efforts are being made to actually build such quantum devices. One of the main challenges is that quantum systems are inherently noisy. However, once the noise is sufficiently suppressed, one can use quantum error correction to suppress arbitrarily by storing the relevant information in a redundant fashion. This course introduces the theoretical basics for this task. An emphasize will be put on conceptual and mathematical aspects. <i>Skills</i> After completing this module, students are able to • reproduce the knowledge taught in the course, • reproduce simpler proofs of the course and reproduce the ideas of the more complicated ones, • establish connections between the concepts taught, and • apply the learned knowledge to concrete problems.			
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>				
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	15 %	Exercises	
Examination	Written exam			
Examination duration and scale	120 min			
Assignment for the Following Curricula	Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Microelectronics and Microsystems: Specialisation Engineering for Quantum Technologies: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L3249: Advanced Quantum Computing	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Martin Kliesch
Language	EN
Cycle	SoSe
Content	• Concept of quantum channels as completely positive and trace-preserving maps • Basic understanding of what error can be corrected (quantum error-correction conditions) • General stabilizer codes and examples • The Clifford group and the Gottesmann-Knill theorem • Transversal gate sets and the Eastin-Knill theorem • The threshold theorem and basic ideas of fault tolerant quantum computation
Literature	Main reference: D. Gottesman, Surviving as a quantum computer in a classical world (2024), book in press, pdf available online.

Course L3250: Advanced Quantum Computing	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Martin Kliesch
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M2072: Information Geometry				
Courses				
Title			Typ	
Information Geometry (L3383)			Lecture	2
Information Geometry (L3384)			Recitation Section (small)	3
Module Responsible	Prof. Nihat Ay			
Admission Requirements	None			
Recommended Previous Knowledge	Linear Algebra, Analysis, Fundamentals of Probability Theory, Basic Programming Course			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	The students know: - statistical models - geometric concepts for statistical models - spherical geometry - Fisher-Rao metric and Fisher information - Shannon entropy - maximum entropy method - exponential family - Kullback-Leibler divergence and information projections - statistical estimators and Cramér-Rao inequality - Fisher efficiency of statistical estimators - Maximum-Likelihood method - the natural gradient for learning algorithms - graphical models and probabilistic neural networks - Gibbs-sampling and generative models			
<i>Skills</i>	The students are able to: - apply geometric concepts in the context of probability theory - evaluate geometric quantities in the context of probability theory - apply the maximum entropy method - apply efficient statistical estimation - derive and apply learning algorithms based on the natural gradient - train generative neural networks			
Personal Competence				
<i>Social Competence</i>	Students are able to work on complex problems both independently and in teams. They can exchange ideas with each other and use their individual strengths to solve problems.			
<i>Autonomy</i>	Students are able to independently investigate a complex problem and assess which competencies are required to solve it.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	None			
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Core Qualification: Compulsory			

Course L3383: Information Geometry	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Nihat Ay
Language	EN
Cycle	SoSe
Content	In this lecture, unifying and natural geometric concepts for probability theory are developed and formalized. The approach is designed in a suggestive manner and illustrated using elementary examples. The application of the resulting formalism is demonstrated using examples in the context of statistics, the theory of graphical models, and machine learning. Specific contents: - statistical models - geometric concepts for statistical models - spherical geometry - Fisher-Rao metric and Fisher information - Shannon entropy - maximum entropy method - exponential family - Kullback-Leibler divergence and information projections - statistical estimators and Cramér-Rao inequality - Fisher efficiency of statistical estimators - Maximum-Likelihood method - the natural gradient for learning algorithms - graphical models and probabilistic neural networks - Gibbs-sampling and generative models
Literature	- S. Amari, H. Nagaoka: Methods of Information Geometry. Oxford University Press, 2000. - S. Amari: Information Geometry and Its Applications. Springer, 2016. - N. Ay, J. Jost, L. Schwachhöfer, H. V. Lê. Information Geometry, 2017. - S. Lauritzen: Graphical Models. Oxford University Press, 1996 (reprinted 2004). - J. Pearl: Causality: Models, Reasoning and Inference. Second edition, Cambridge University Press, 2009.

Course L3384: Information Geometry	
Typ	Recitation Section (small)
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Nihat Ay, Adwait Datar
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1552: Advanced Machine Learning			
Courses			
Title	Typ	Hrs/wk	CP
Advanced Machine Learning (L2322)	Lecture	2	3
Advanced Machine Learning (L2323)	Recitation Section (small)	2	3
Module Responsible	Dr. Jens-Peter Zemke		
Admission Requirements	None		
Recommended Previous Knowledge	1. Mathematics I-III 2. Numerical Mathematics 1/ Numerics 3. Programming skills, preferably in Python		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	<i>Knowledge</i> Students are able to name, state and classify state-of-the-art neural networks and their corresponding mathematical basics. They can assess the difficulties of different neural networks. <i>Skills</i> Students are able to implement, understand, and, tailored to the field of application, apply neural networks. <i>Social Competence</i> Students can • develop and document joint solutions in small teams; • form groups to further develop the ideas and transfer them to other areas of applicability; • form a team to develop, build, and advance a software library. <i>Autonomy</i> Students are able to • correctly assess the time and effort of self-defined work; • assess whether the supporting theoretical and practical exercises are better solved individually or in a team; • define test problems for testing and expanding the methods; • assess their individual progress and, if necessary, to ask questions and seek help.		
<i>Knowledge</i>			
<i>Skills</i>			
<i>Social Competence</i>			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	90 min		
Assignment for the Following Curricula	Computational Engineering: Core Qualification: Elective Compulsory Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Core Qualification: Compulsory Computer Science in Engineering: Specialisation III. Mathematics: Elective Compulsory Interdisciplinary Mathematics: Specialisation Computational Methods in Biomedical Imaging: Compulsory Mechatronics: Core Qualification: Elective Compulsory Technomathematics: Specialisation I. Mathematics: Elective Compulsory Technomathematics: Specialisation II. Informatics: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory		

Course L2322: Advanced Machine Learning	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Jens-Peter Zemke
Language	EN
Cycle	WiSe
Content	1. Basics: analogy; layout of neural nets, universal approximation, NP-completeness 2. Feedforward Neural Networks: backpropagation, variants of Stochastic Gradients 3. Convolutional Neural Networks: idea, layout, FFT and Winograds algorithms, implementation details 4. Adversarial Attacks 5. Recurrent Neural Networks: idea, dynamical systems, training, LSTM 6. Residual Neural Networks 7. Neural ODEs 8. Autoencoder and Generative Adversarial Networks 9. Attention and Transformers
Literature	1. Skript 2. Online-Werke: ◦ http://neuralnetworksanddeeplearning.com/ ◦ https://www.deeplearningbook.org/

Course L2323: Advanced Machine Learning	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Jens-Peter Zemke
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1964: Advanced Complexity Theory			
Courses			
Title	Typ	Hrs/wk	CP
Advanced Complexity Theory (L3207)	Lecture	2	3
Advanced Complexity Theory (L3208)	Recitation Section (small)	2	3
Module Responsible	Prof. Antoine Wiehe		
Admission Requirements	None		
Recommended Previous Knowledge	Knowledge from the modules Automata Theory and Formal Languages as well as Computability and Complexity Theory.		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i>	Students know - the different models of computation: Turing machines (deterministic, non-deterministic, probabilistic) and boolean circuits - the definition of standard complexity classes and how to organize them in a hierarchy; - the fundamental hierarchy theorems (deterministic time hierarchy, non-deterministic time hierarchy, space hierarchy);		
<i>Skills</i>	The student can apply his knowledge of the standard complexity classes to reason about new models of computation.		
Personal Competence <i>Social Competence</i> <i>Autonomy</i>			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Oral exam		
Examination duration and scale	40 min		
Assignment for the Following Curricula	Computer Science: Specialisation III. Mathematics: Elective Compulsory		

Course L3207: Advanced Complexity Theory	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Antoine Wiehe
Language	EN
Cycle	WiSe
Content	Computational complexity is a field from theoretical computer science that is concerned with the study of computational problems and their organisation in various classes corresponding to the amount of resources (like time or memory) that are needed to solve the problems. This is one of the most active research fields in theoretical computer science and a number of famous open problems are directly connected to computational complexity (for example, the Millennium problem "P vs. NP" or the complexity of the graph isomorphism problem). The course will cover the core and advanced material from this discipline, such as the important complexity classes (including, but not limited to, P and NP), as well as the classical results relating these classes.
Literature	- Computational complexity: a modern approach, S. Arora and B. Barak - Computational complexity, C. H. Papadimitriou

Course L3208: Advanced Complexity Theory	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Antoine Wiehe
Language	EN
Cycle	WiSe
Content	Computational complexity is a field from theoretical computer science that is concerned with the study of computational problems and their organisation in various classes corresponding to the amount of resources (like time or memory) that are needed to solve the problems. This is one of the most active research fields in theoretical computer science and a number of famous open problems are directly connected to computational complexity (for example, the Millennium problem "P vs. NP" or the complexity of the graph isomorphism problem). The course will cover the core and advanced material from this discipline, such as the important complexity classes (including, but not limited to, P and NP), as well as the classical results relating these classes.
Literature	- Computational complexity: a modern approach, S. Arora and B. Barak - Computational complexity, C. H. Papadimitriou

Module M1966: Mathematical Image Processing			
Courses			
Title	Typ	Hrs/wk	CP
Mathematical Image Processing (L0991)	Lecture	3	4
Mathematical Image Processing (L0992)	Recitation Section (small)	1	2
Module Responsible	Prof. Marko Lindner		
Admission Requirements	None		
Recommended Previous Knowledge	- Analysis: partial derivatives, gradient, directional derivative - Linear Algebra: eigenvalues, least squares solution of a linear system		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i>	Students are able to - characterize and compare diffusion equations - explain elementary methods of image processing - explain methods of image segmentation and registration - sketch and interrelate basic concepts of functional analysis		
<i>Skills</i>	Students are able to - implement and apply elementary methods of image processing - explain and apply modern methods of image processing		
Personal Competence <i>Social Competence</i>	Students are able to work together in heterogeneously composed teams (i.e., teams from different study programs and background knowledge) and to explain theoretical foundations.		
<i>Autonomy</i>	- Students are capable of checking their understanding of complex concepts on their own. They can specify open questions precisely and know where to get help in solving them. - Students have developed sufficient persistence to be able to work for longer periods in a goal-oriented manner on hard problems.		
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	20 min		
Assignment for the Following Curricula	Bioprocess Engineering: Specialisation A - General Bioprocess Engineering: Elective Compulsory Computer Science: Specialisation III. Mathematics: Elective Compulsory Computer Science in Engineering: Specialisation III. Mathematics: Elective Compulsory Interdisciplinary Mathematics: Specialisation Computational Methods in Biomedical Imaging: Compulsory Mechatronics: Core Qualification: Elective Compulsory Technomathematics: Specialisation I. Mathematics: Elective Compulsory Technomathematics: Specialisation II. Informatics: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory Process Engineering: Specialisation Process Engineering: Elective Compulsory		

Course L0991: Mathematical Image Processing	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Marko Lindner
Language	DE/EN
Cycle	WiSe
Content	- basic methods of image processing - smoothing filters - the diffusion / heat equation - variational formulations in image processing - edge detection - de-convolution - inpainting - image segmentation - image registration
Literature	Bredies/Lorenz: Mathematische Bildverarbeitung

Course L0992: Mathematical Image Processing	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Marko Lindner
Language	DE/EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1973: Numerical Methods for Partial Differential Equations				
Courses				
Title		Typ	Hrs/wk	CP
Numerics of Partial Differential Equations (L1247)		Lecture	2	3
Numerics of Partial Differential Equations (L1248)		Recitation Section (small)	2	3
Module Responsible	Prof. Daniel Ruprecht			
Admission Requirements	None			
Recommended Previous Knowledge	- Mathematik I - IV (for Engineering Students) or Analysis & Linear Algebra I + II for Technomathematicians - Numerical mathematics 1 - Numerical methods for ordinary differential equations - Basic knowledge of MATLAB, Python or a similar programming language.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i>	- Students can classify partial differential equations according to the three basic types. - They know typical numerical methods like finite differences or finite volumes. - Students know the theoretical convergence results and other important properties of these methods.			
<i>Skills</i>	Students are capable of formulating solution strategies for given partial differential equations, can comment on theoretical properties regarding convergence and are able to implement and test these methods.			
Personal Competence <i>Social Competence</i>	Students are able of working together in heterogeneous teams (i.e., teams from different study programs and background knowledge) and to explain theoretical foundations.			
<i>Autonomy</i>	- Students are capable of checking their understanding of complex concepts on their own. They can specify open questions precisely and know where to get help in solving them. - Students have developed sufficient mental stamina to work on hard problems for an extended period of time			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	None	Excercises	Regelmässige Bearbeitung von Übungsaufgaben und aktive Teilnahme an den Übungsgruppen
Examination	Oral exam			
Examination duration and scale	30 min			
Assignment for the Following Curricula	Computational Engineering: Core Qualification: Elective Compulsory Computer Science: Specialisation III. Mathematics: Elective Compulsory Technomathematics: Specialisation I. Mathematics: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Simulation Technology: Elective Compulsory			

Course L1247: Numerics of Partial Differential Equations	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Daniel Ruprecht
Language	DE/EN
Cycle	WiSe
Content	Elementary Theory and Numerics of PDEs - types of PDEs - well posed problems - finite differences - finite volumes - applications
Literature	Dale R. Durran: Numerical Methods for Fluid Dynamics. Randall J. LeVeque: Numerical Methods for Conservation Laws.

Course L1248: Numerics of Partial Differential Equations	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Daniel Ruprecht
Language	DE/EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Specialization IV. Subject Specific Focus

Module M1565: Technical Complementary Course I for CSMS
Courses

Title	Typ	Hrs/wk	CP
Module Responsible	Dozenten des SD E		
Admission Requirements	None		
Recommended Previous Knowledge			
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i> <i>Skills</i> Personal Competence <i>Social Competence</i> <i>Autonomy</i>			
Workload in Hours	Depends on choice of courses		
Credit points	6		
Assignment for the Following Curricula	Computer Science: Specialisation IV. Subject Specific Focus: Elective Compulsory		

Module M1566: Technical Complementary Course II for CSMS				
Courses				
Title	Typ		Hrs/wk	CP
Module Responsible	Dozenten des SD E			
Admission Requirements	None			
Recommended Previous Knowledge				
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i> <i>Skills</i> Personal Competence <i>Social Competence</i> <i>Autonomy</i>				
Workload in Hours	Depends on choice of courses			
Credit points	6			
Assignment for the Following Curricula	Computer Science: Specialisation IV. Subject Specific Focus: Elective Compulsory			

Module M2115: Advanced Seminar EIM I			
Courses			
Title	Typ	Hrs/wk	CP
Advanced Seminar EIM I (L3385)	Seminar	2	3
Module Responsible	Dozenten des SD E		
Admission Requirements	None		
Recommended Previous Knowledge	Basic knowledge in the area of the seminar within the fields of computer science, communication technology, electrical engineering or mathematics at the Master's level.		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	<i>Knowledge</i> The students are able to • explicate a specific topic from the field of the seminar, • describe complex issues, • present different views and evaluate in a critical way. <i>Skills</i> The students are able to • familiarize in a specific topic in limited time, • realize a literature survey on the specific topic and cite in a correct way, • elaborate a presentation and give a lecture to a selected audience, • sum up the presentation in 10-15 lines, • answer questions in the final discussion. Personal Competence <i>Social Competence</i> The students are able to • elaborate and introduce a topic for a certain audience, • discuss the topic, content and structure of the presentation with the instructor, • discuss certain aspects with the audience, and • as the lecturer listen and respond to questions from the audience. <i>Autonomy</i> The students are able to • define the task in question in an autonomous way, • develop the necessary knowledge, • use appropriate work equipment, and • guided by an instructor critically check the working status.		
Workload in Hours	Depends on choice of courses		
Credit points	3		
Assignment for the Following Curricula	Computer Science: Specialisation IV. Subject Specific Focus: Elective Compulsory Data Science: Core Qualification: Compulsory Electrical Engineering and Information Technology: Specialisation Microwave Engineering, Optics, and Electromagnetic Compatibility: Compulsory Electrical Engineering and Information Technology: Specialisation Medical Technology: Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Compulsory Electrical Engineering and Information Technology: Specialisation Nanoelectronics and Microsystems Technology: Compulsory Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Compulsory Electrical Engineering and Information Technology: Specialisation Wireless and Sensor Technologies: Compulsory Computer Science in Engineering: Specialisation IV. Subject Specific Focus: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Elective Compulsory		

Course L3385: Advanced Seminar EIM I	
Typ	Seminar
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Fachtheoretisch-fachpraktische Arbeit
Examination duration and scale	x
Lecturer	Dozenten des SD E
Language	EN
Cycle	WiSe/SoSe
Content	- Seminar presentations by enrolled students about selected topics of computer science, communication technology, electrical engineering or mathematics - Active participation in discussions
Literature	Wird vom Veranstalter bekanntgegeben.

Module M2116: Advanced Seminar EIM II				
Courses				
Title	Typ		Hrs/wk	CP
Advanced Seminar EIM II (L3386)	Seminar		2	3
Module Responsible	Dozenten des SD E			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge in the area of the seminar within the fields of computer science, communication technology, electrical engineering or mathematics at the Master's level.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> The students are able to - explicate a specific topic from the field of the seminar, - describe complex issues, - present different views and evaluate in a critical way. <i>Skills</i> The students are able to - familiarize in a specific topic in limited time, - realize a literature survey on the specific topic and cite in a correct way, - elaborate a presentation and give a lecture to a selected audience, - sum up the presentation in 10-15 lines, - answer questions in the final discussion. Personal Competence <i>Social Competence</i> The students are able to - elaborate and introduce a topic for a certain audience, - discuss the topic, content and structure of the presentation with the instructor, - discuss certain aspects with the audience, and - as the lecturer listen and respond to questions from the audience. <i>Autonomy</i> The students are able to - define the task in question in an autonomous way, - develop the necessary knowledge, - use appropriate work equipment, and - guided by an instructor critically check the working status.			
Workload in Hours	Depends on choice of courses			
Credit points	3			
Assignment for the Following Curricula	Computer Science: Specialisation IV. Subject Specific Focus: Elective Compulsory Computer Science in Engineering: Specialisation IV. Subject Specific Focus: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Elective Compulsory			

Course L3386: Advanced Seminar EIM II	
Typ	Seminar
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Fachtheoretisch-fachpraktische Arbeit
Examination duration and scale	x
Lecturer	Dozenten des SD E
Language	EN
Cycle	WiSe/SoSe
Content	- Seminar presentations by enrolled students about selected topics of computer science, communication technology, electrical engineering or mathematics - Active participation in discussions
Literature	Wird vom Veranstalter bekanntgegeben.

Thesis

Module M-002: Master Thesis					
Courses					
Title		Typ	Hrs/wk	CP	
Module Responsible	Professoren der TUHH				
Admission Requirements	According to General Regulations §21 (1): At least 60 credit points have to be achieved in study programme. The examinations board decides on exceptions.				
Recommended Previous Knowledge					
Educational Objectives	After taking part successfully, students have reached the following learning results				
Professional Competence	- The students can use specialized knowledge (facts, theories, and methods) of their subject competently on specialized issues. - The students can explain in depth the relevant approaches and terminologies in one or more areas of their subject, describing current developments and taking up a critical position on them. - The students can place a research task in their subject area in its context and describe and critically assess the state of research.				
Knowledge					
Skills					The students are able: - To select, apply and, if necessary, develop further methods that are suitable for solving the specialized problem in question. - To apply knowledge they have acquired and methods they have learnt in the course of their studies to complex and/or incompletely defined problems in a solution-oriented way. - To develop new scientific findings in their subject area and subject them to a critical assessment.
Personal Competence					
Social Competence	Students can - Both in writing and orally outline a scientific issue for an expert audience accurately, understandably and in a structured way. - Deal with issues competently in an expert discussion and answer them in a manner that is appropriate to the addressees while upholding their own assessments and viewpoints convincingly.				
Autonomy	Students are able: - To structure a project of their own in work packages and to work them off accordingly. - To work their way in depth into a largely unknown subject and to access the information required for them to do so. - To apply the techniques of scientific work comprehensively in research of their own.				
Workload in Hours	Independent Study Time 900, Study Time in Lecture 0				
Credit points	30				
Course achievement	None				
Examination	Thesis				
Examination duration and scale	According to General Regulations				
Assignment for the Following Curricula	Civil Engineering: Thesis: Compulsory Bioprocess Engineering: Thesis: Compulsory Chemical and Bioprocess Engineering: Thesis: Compulsory Chemical and Bioprocess Engineering: Thesis: Compulsory Computational Engineering: Thesis: Compulsory Computer Science: Thesis: Compulsory Data Science: Thesis: Compulsory Electrical Engineering and Information Technology: Thesis: Compulsory Electrical Engineering: Thesis: Compulsory Energy Systems: Thesis: Compulsory Environmental Engineering: Thesis: Compulsory Aircraft Systems Engineering: Thesis: Compulsory Global Innovation Management: Thesis: Compulsory Computer Science in Engineering: Thesis: Compulsory Information and Communication Systems: Thesis: Compulsory Interdisciplinary Mathematics: Thesis: Compulsory International Production Management: Thesis: Compulsory International Management and Engineering: Thesis: Compulsory Joint European Master in Environmental Studies - Cities and Sustainability: Thesis: Compulsory Logistics, Infrastructure and Mobility: Thesis: Compulsory Aeronautics: Thesis: Compulsory Mechanical Engineering - Product Development and Production: Thesis: Compulsory Materials Science and Engineering: Thesis: Compulsory				

	Materials Science: Thesis: Compulsory
	Mechanical Engineering and Management: Thesis: Compulsory
	Mechatronics: Thesis: Compulsory
	Biomedical Engineering: Thesis: Compulsory
	Microelectronics and Microsystems: Thesis: Compulsory
	Product Development, Materials and Production: Thesis: Compulsory
	Renewable Energies: Thesis: Compulsory
	Naval Architecture and Ocean Engineering: Thesis: Compulsory
	Naval Architecture and Ocean Engineering: Thesis: Compulsory
	Ship and Offshore Technology: Thesis: Compulsory
	Theoretical Mechanical Engineering: Thesis: Compulsory
	Process Engineering: Thesis: Compulsory
	Water and Environmental Engineering: Thesis: Compulsory
	Certification in Engineering & Advisory in Aviation: Thesis: Compulsory